

BFS Traversal

Step 1: Start at E

Legend

Complete

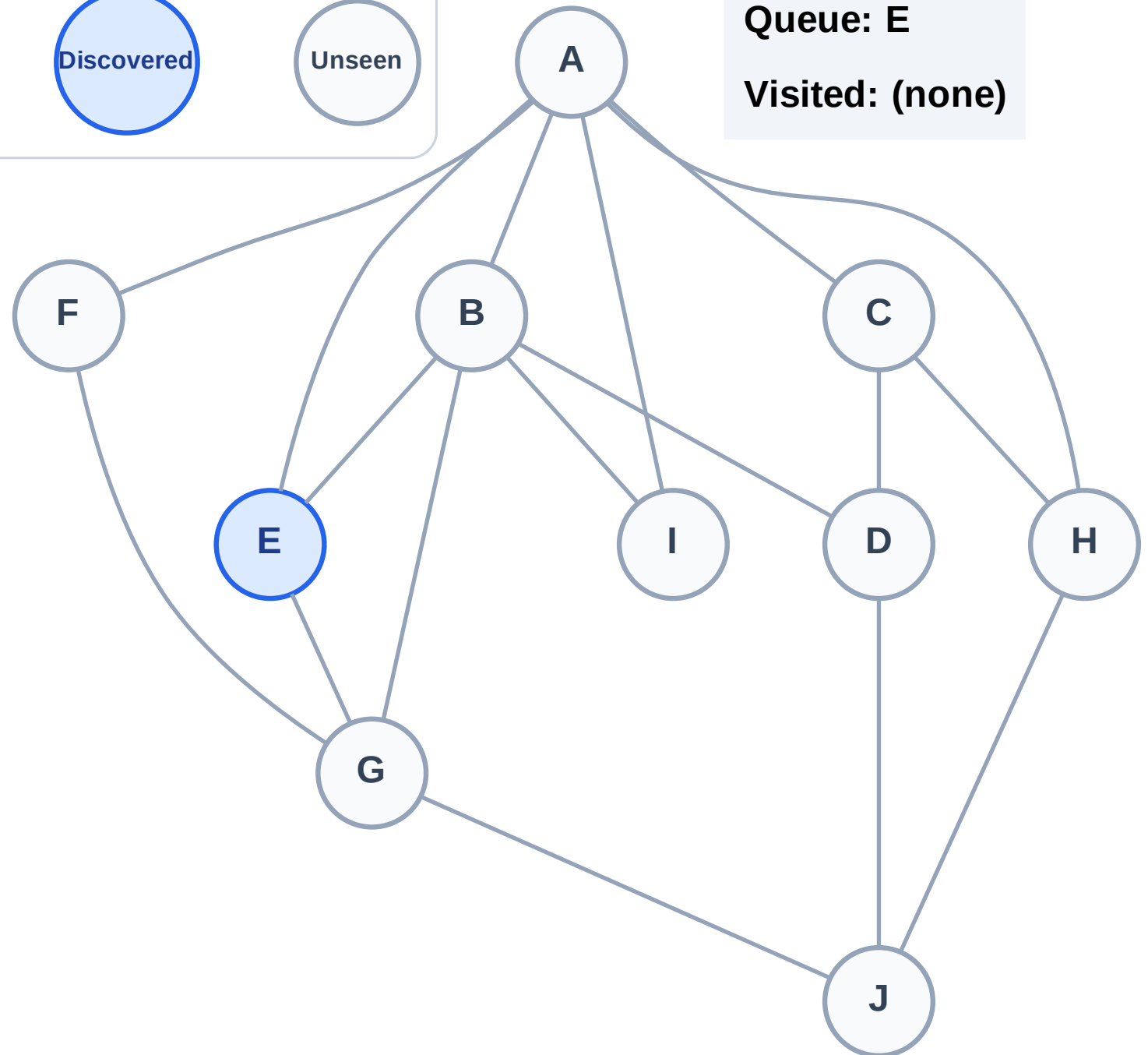
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

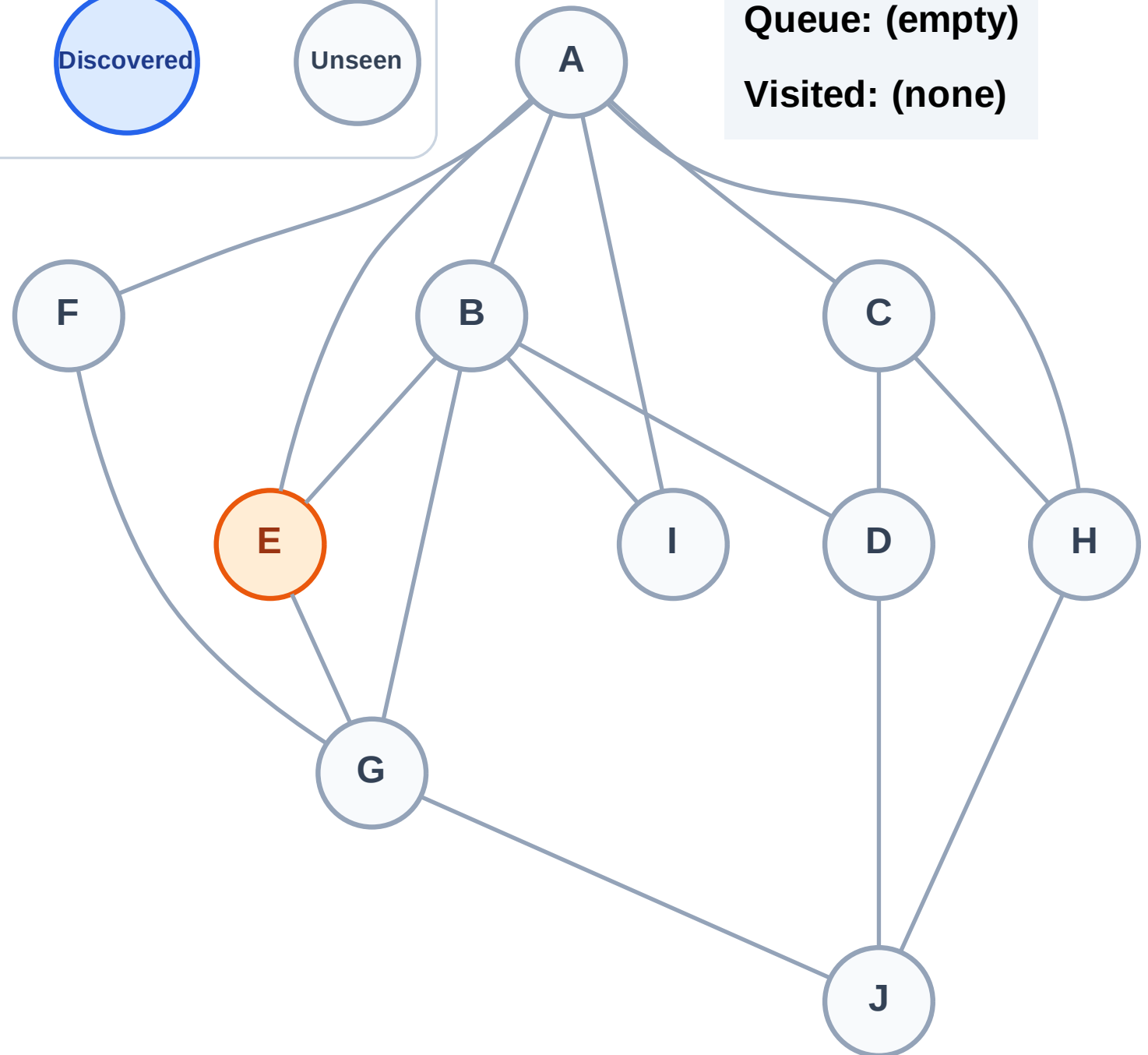
Processing

Discovered

Unseen

Queue: (empty)

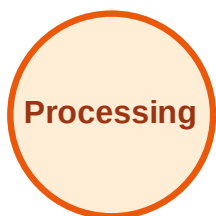
Visited: (none)



BFS Traversal

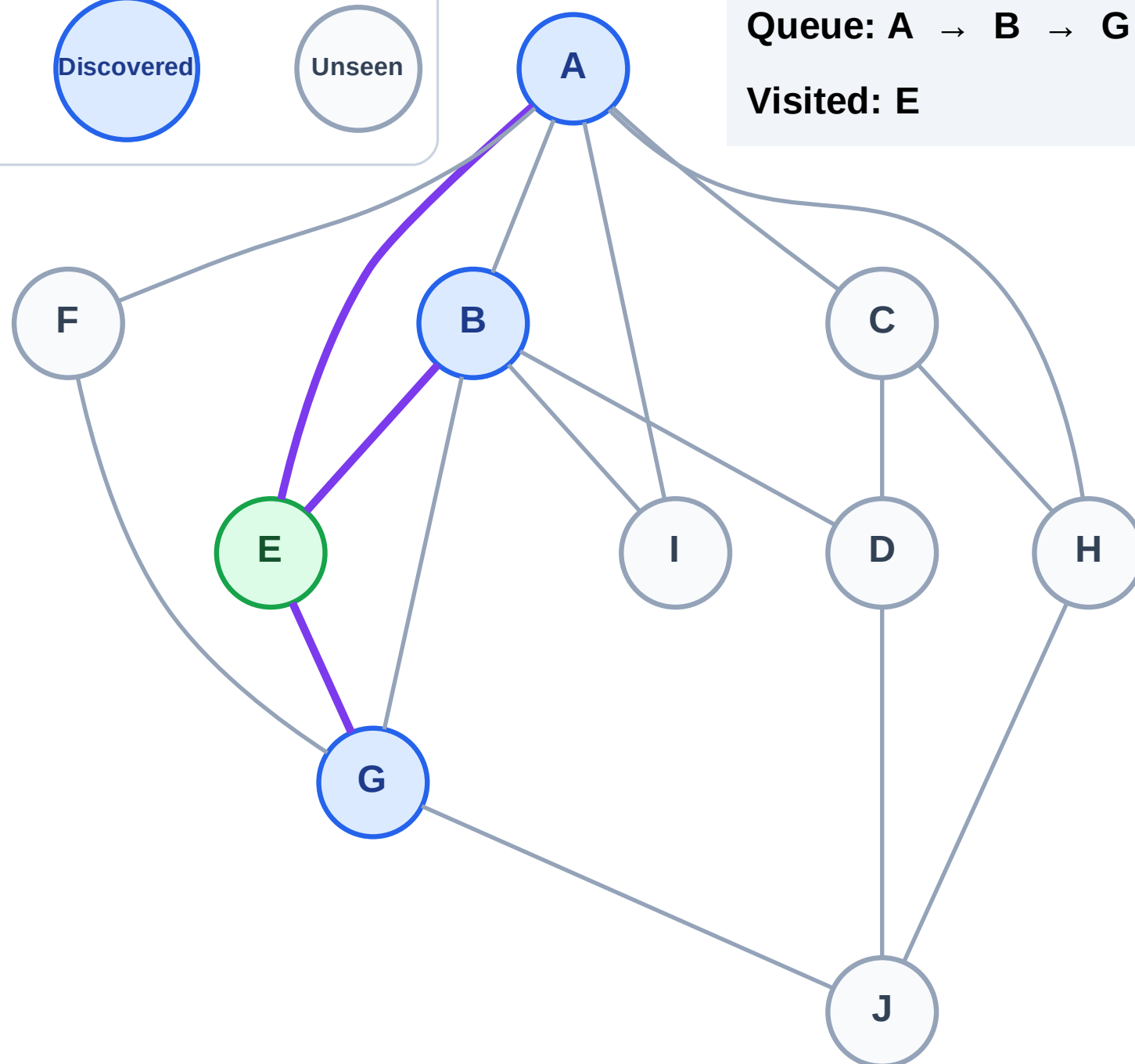
Step 3: Discover A, B, G from E

Legend



Queue: A → B → G

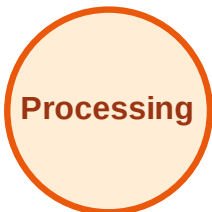
Visited: E



BFS Traversal

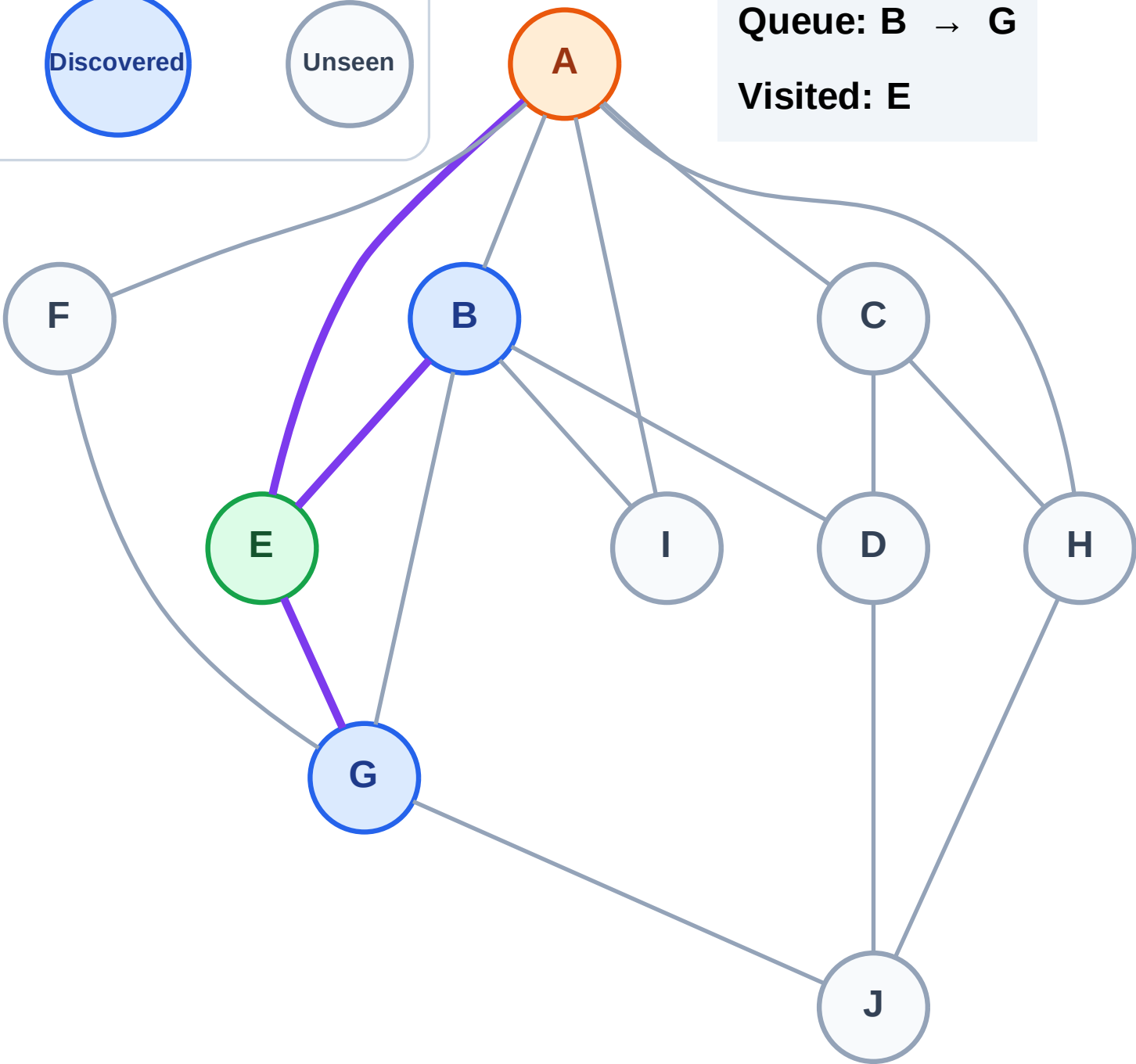
Step 4: Process node A

Legend



Queue: B → G

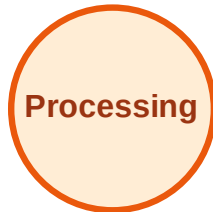
Visited: E



BFS Traversal

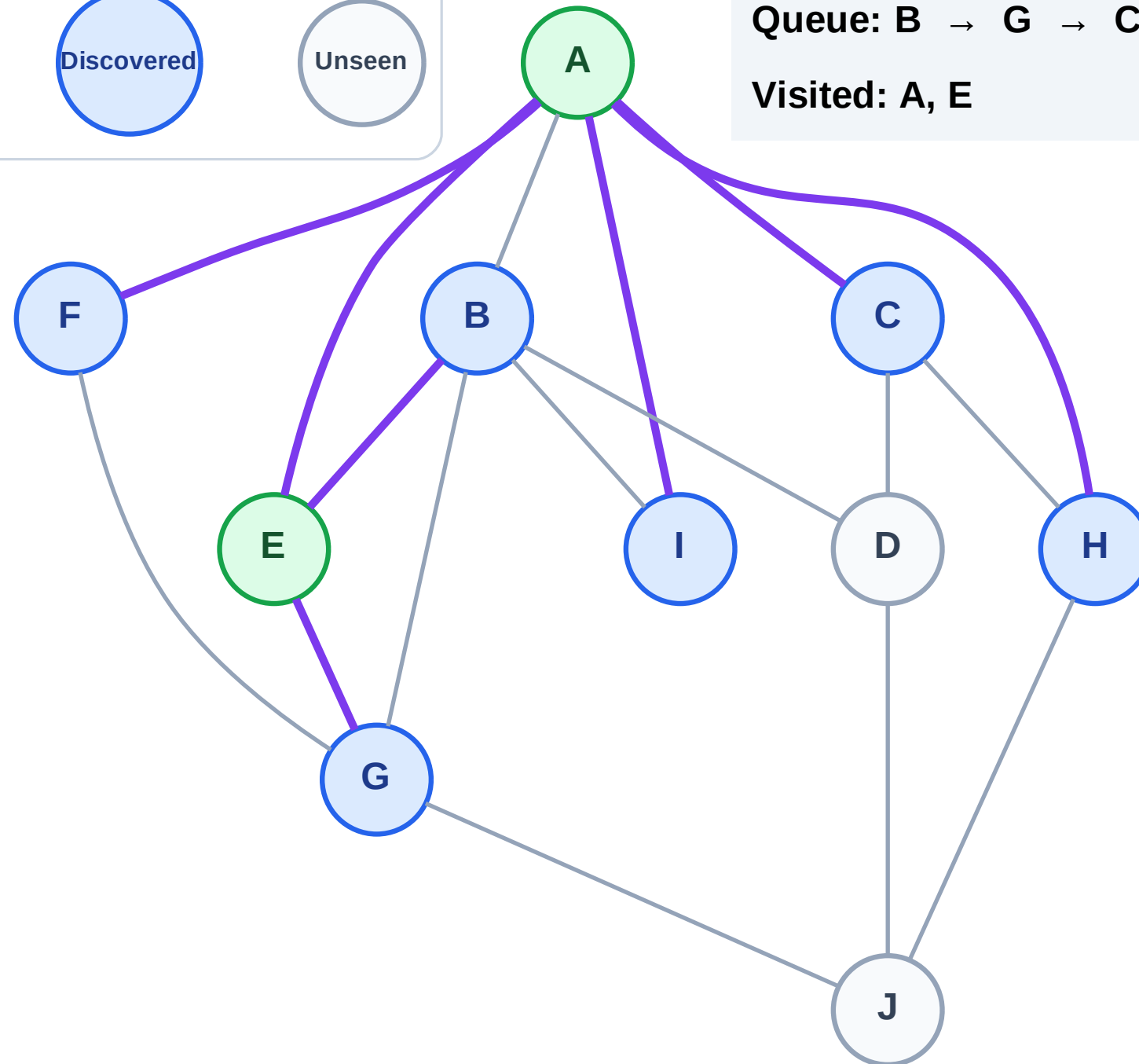
Step 5: Discover C, F, H, I from A

Legend



Queue: B → G → C → F → H → I

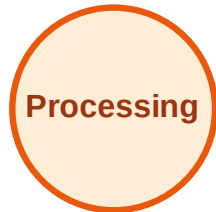
Visited: A, E



BFS Traversal

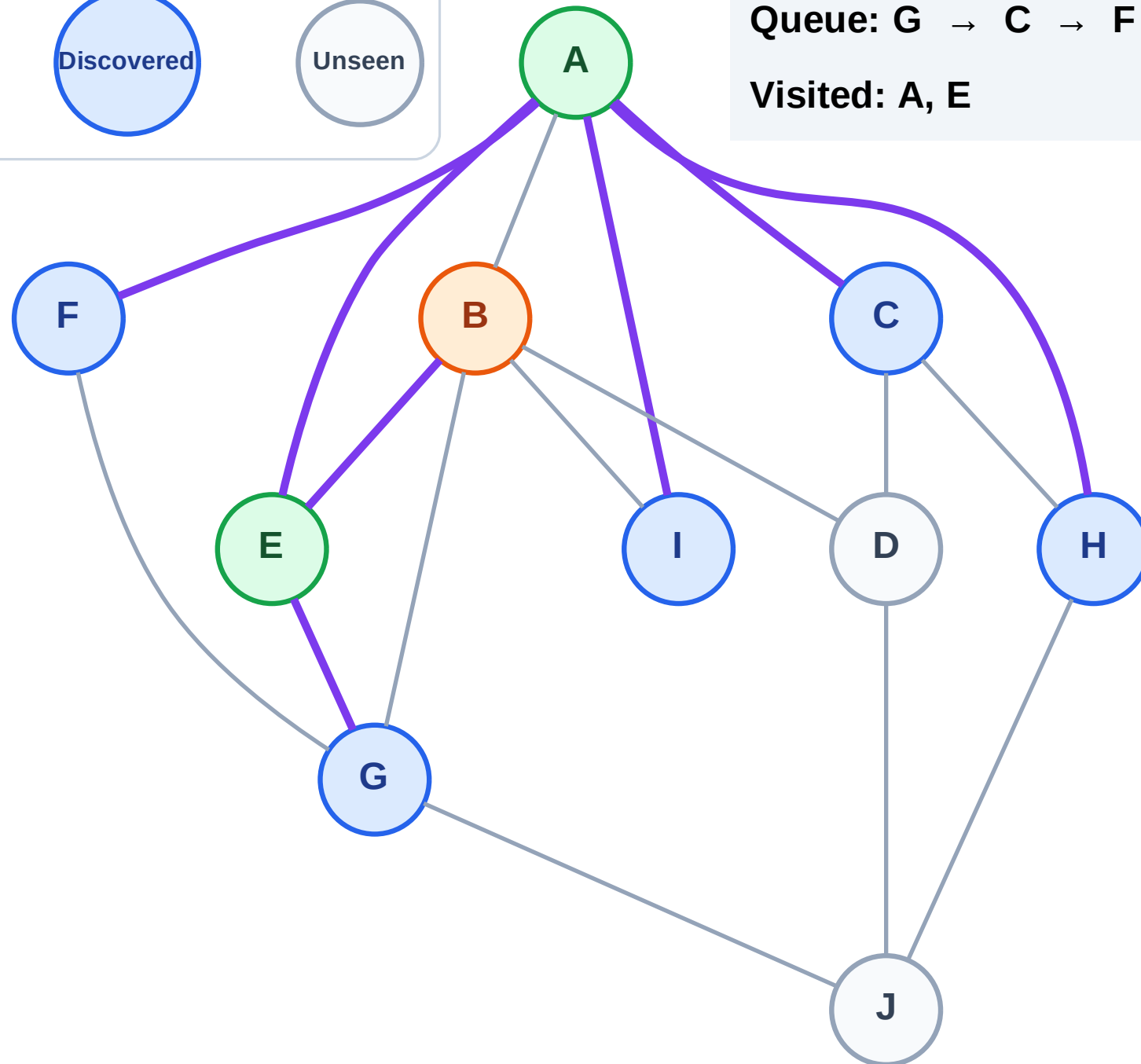
Step 6: Process node B

Legend



Queue: G → C → F → H → I

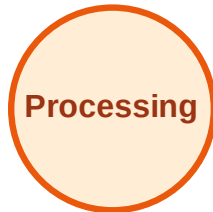
Visited: A, E



BFS Traversal

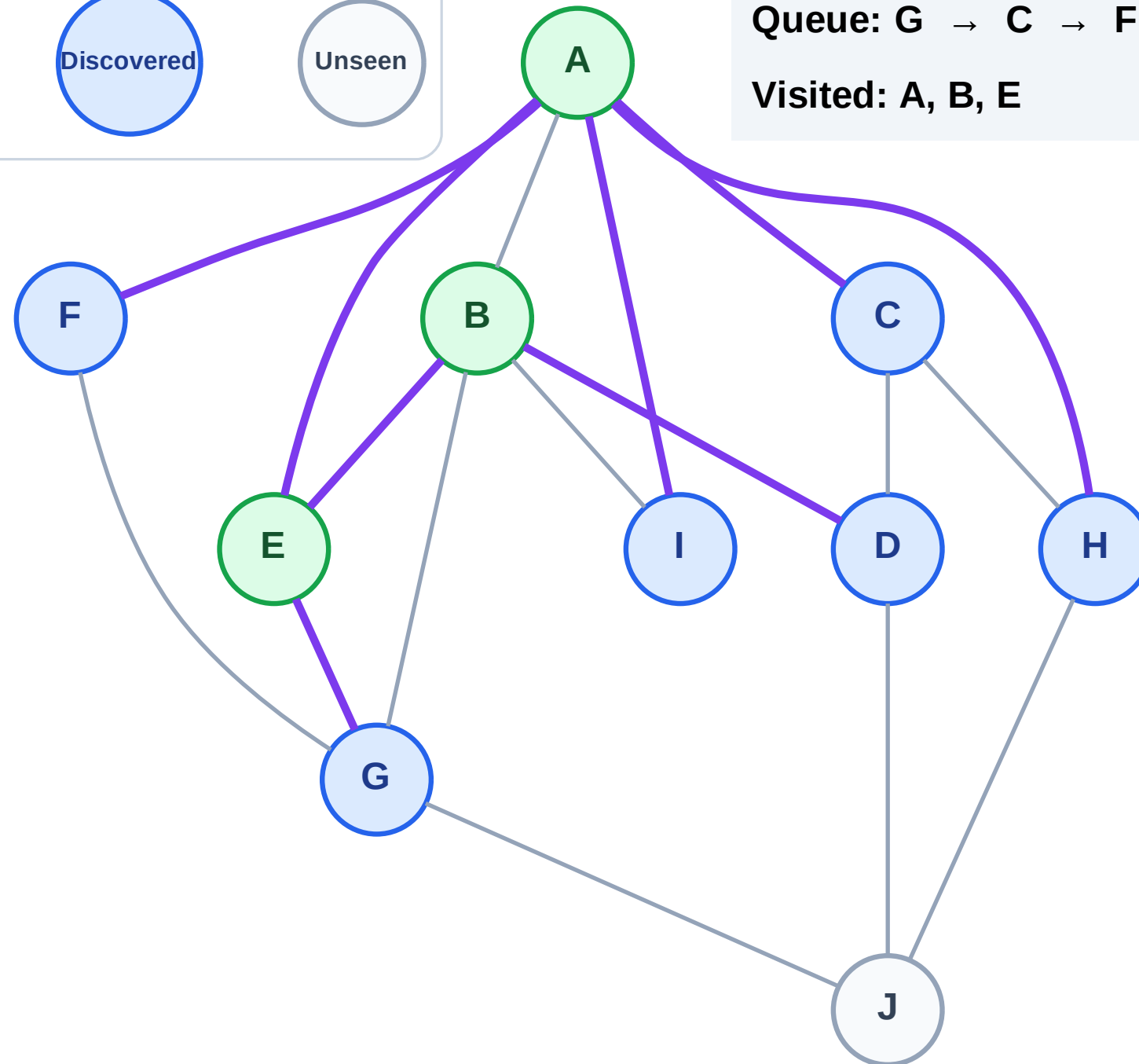
Step 7: Discover D from B

Legend



Queue: G → C → F → H → I → D

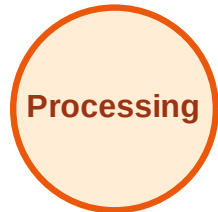
Visited: A, B, E



BFS Traversal

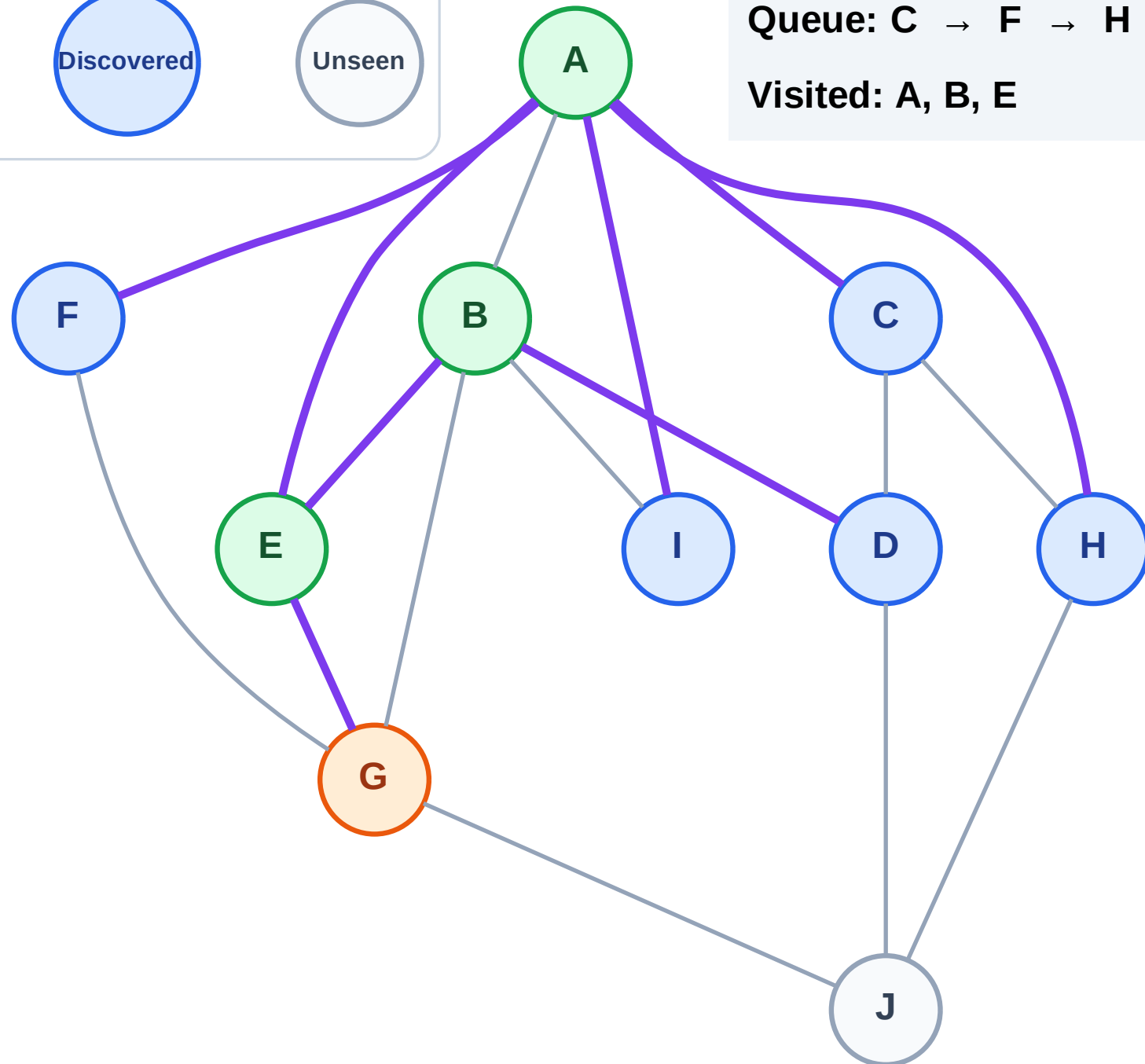
Step 8: Process node G

Legend



Queue: C → F → H → I → D

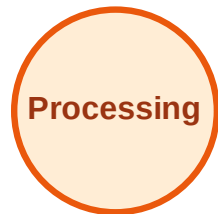
Visited: A, B, E



BFS Traversal

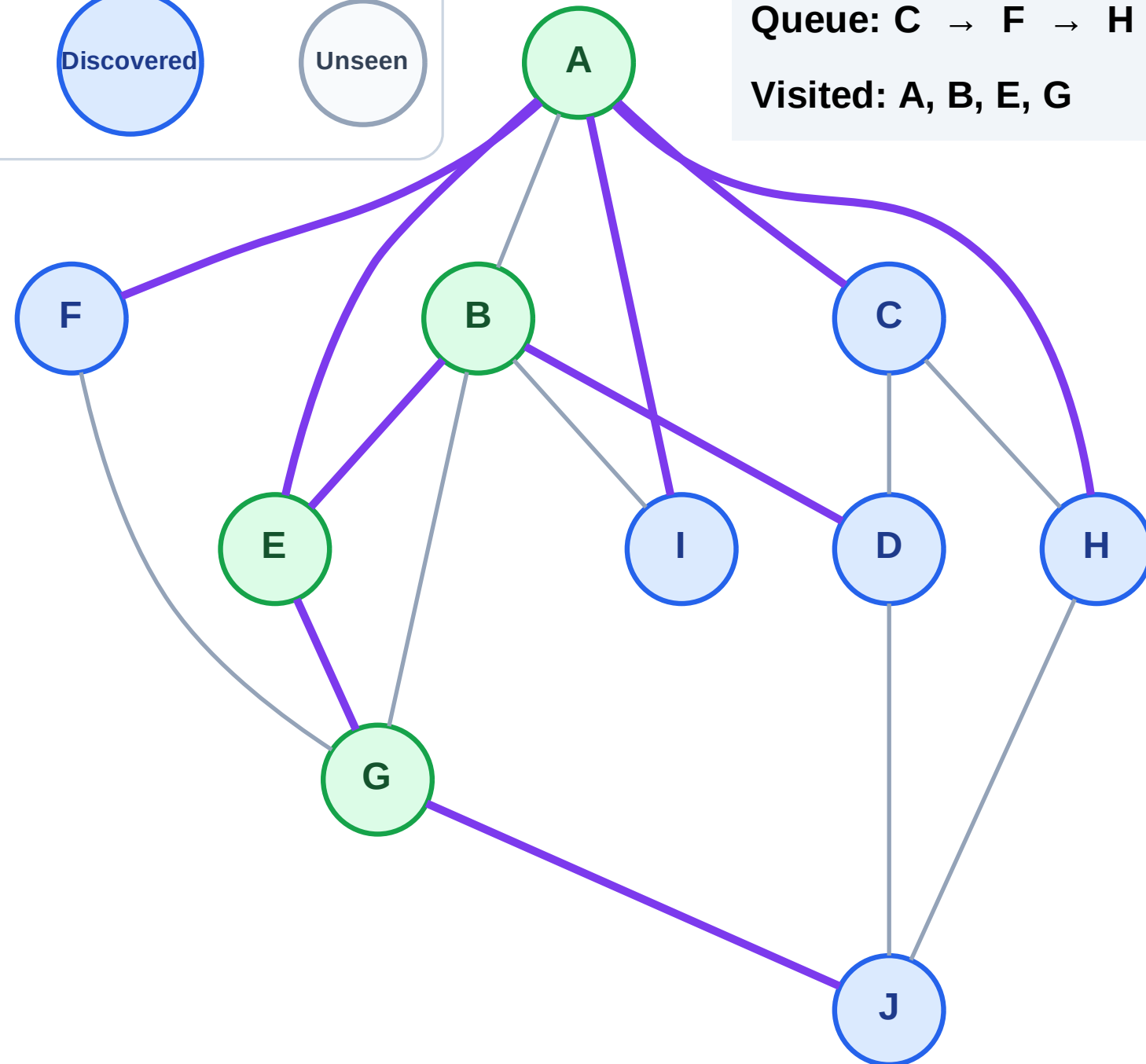
Step 9: Discover J from G

Legend



Queue: C → F → H → I → D → J

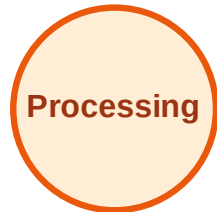
Visited: A, B, E, G



BFS Traversal

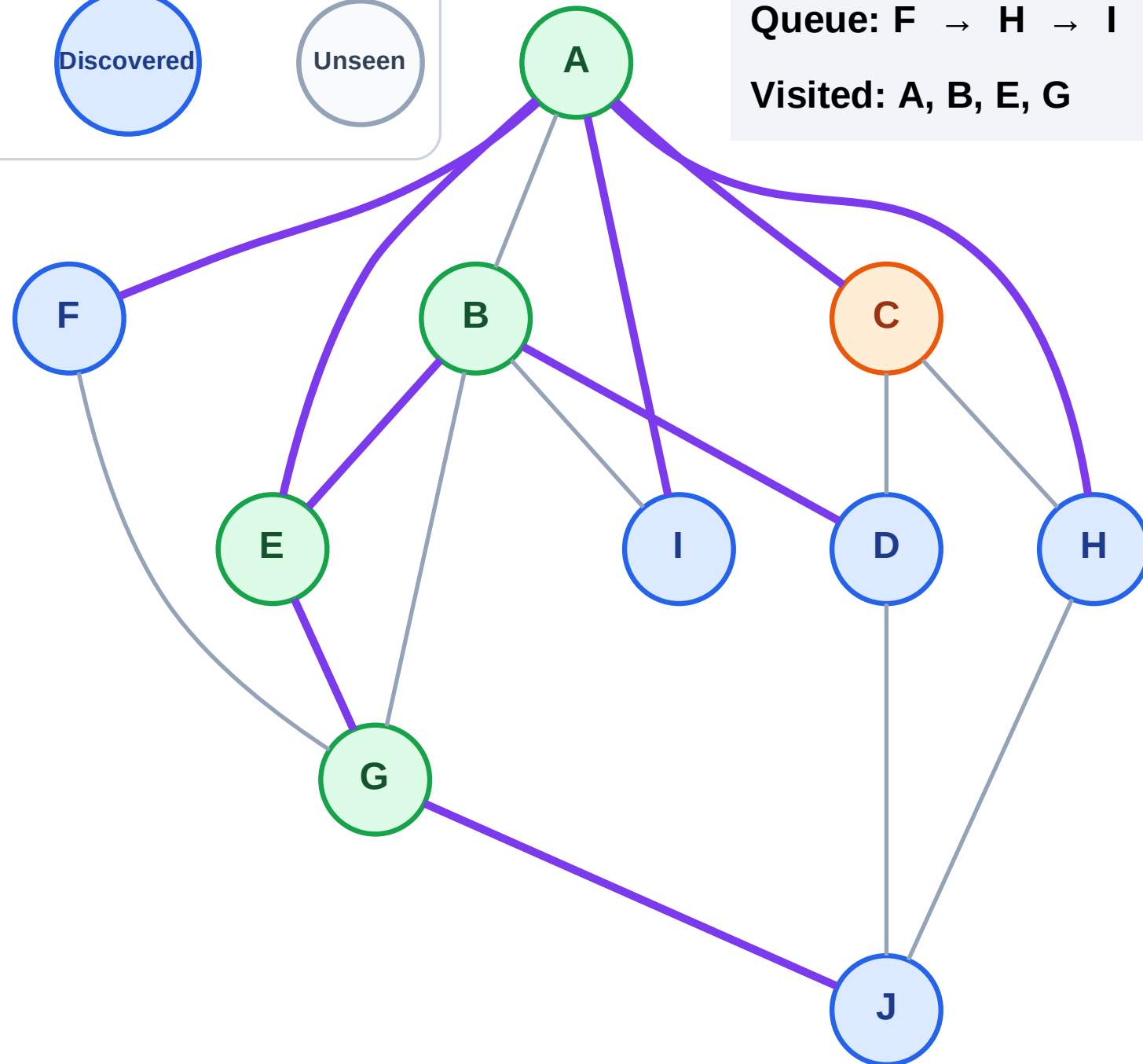
Step 10: Process node C

Legend



Queue: F → H → I → D → J

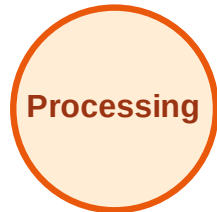
Visited: A, B, E, G



BFS Traversal

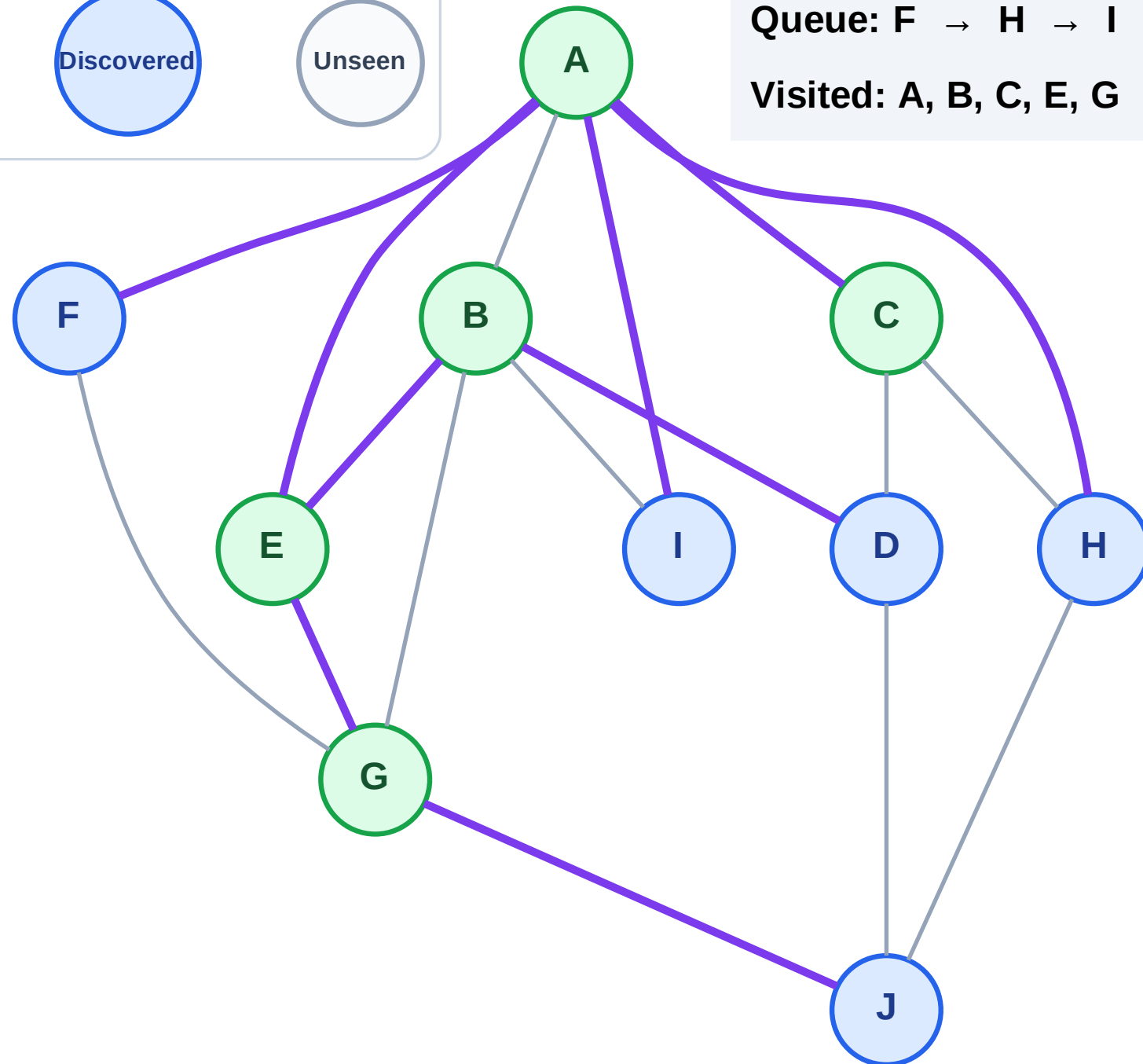
Step 11: No new neighbors from C

Legend



Queue: F → H → I → D → J

Visited: A, B, C, E, G



BFS Traversal

Step 12: Process node F

Legend

Complete

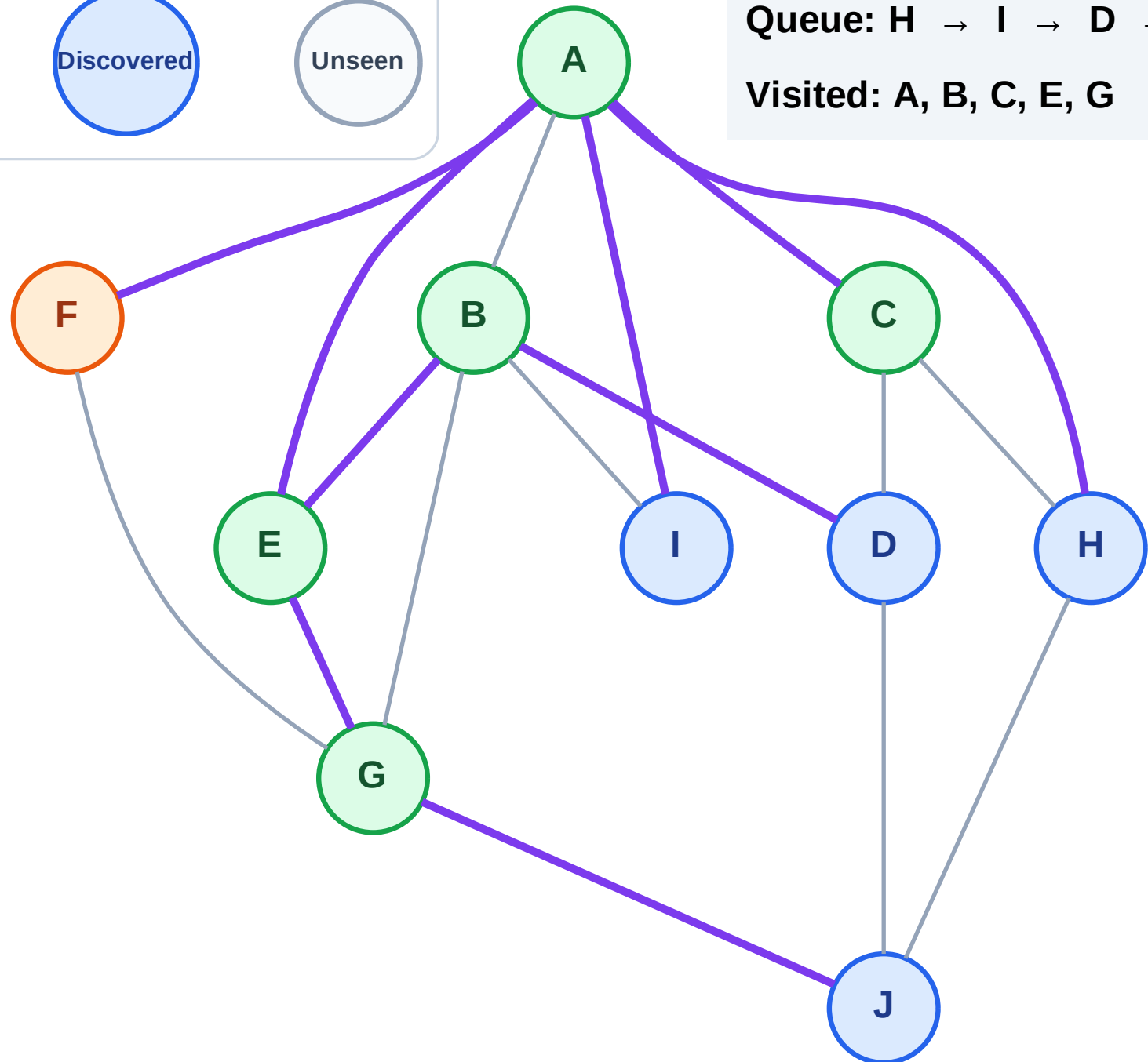
Processing

Discovered

Unseen

Queue: H → I → D → J

Visited: A, B, C, E, G



BFS Traversal

Step 13: No new neighbors from F

Legend

Complete

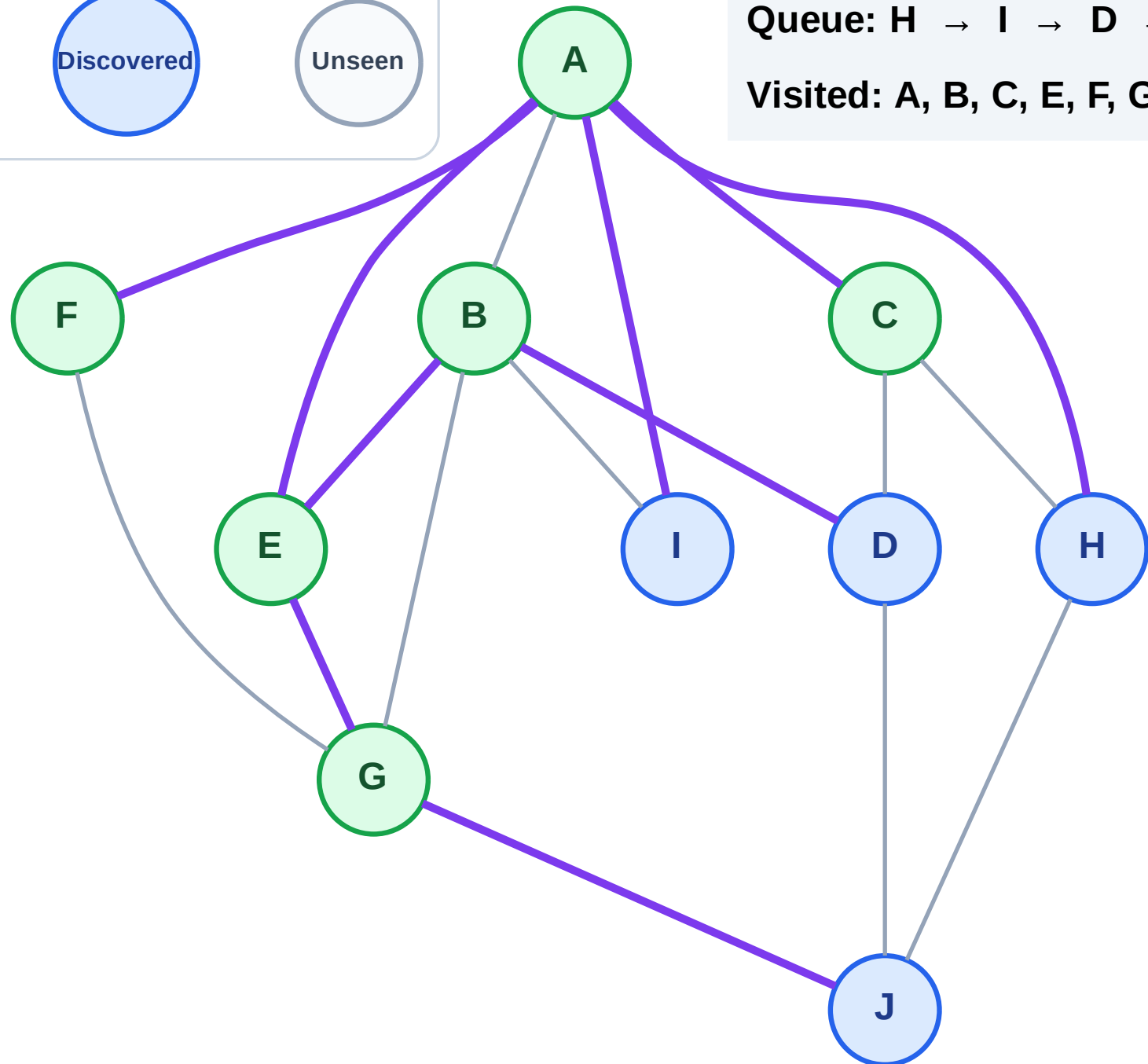
Processing

Discovered

Unseen

Queue: H → I → D → J

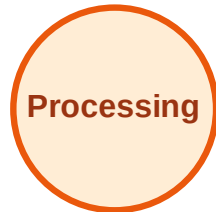
Visited: A, B, C, E, F, G



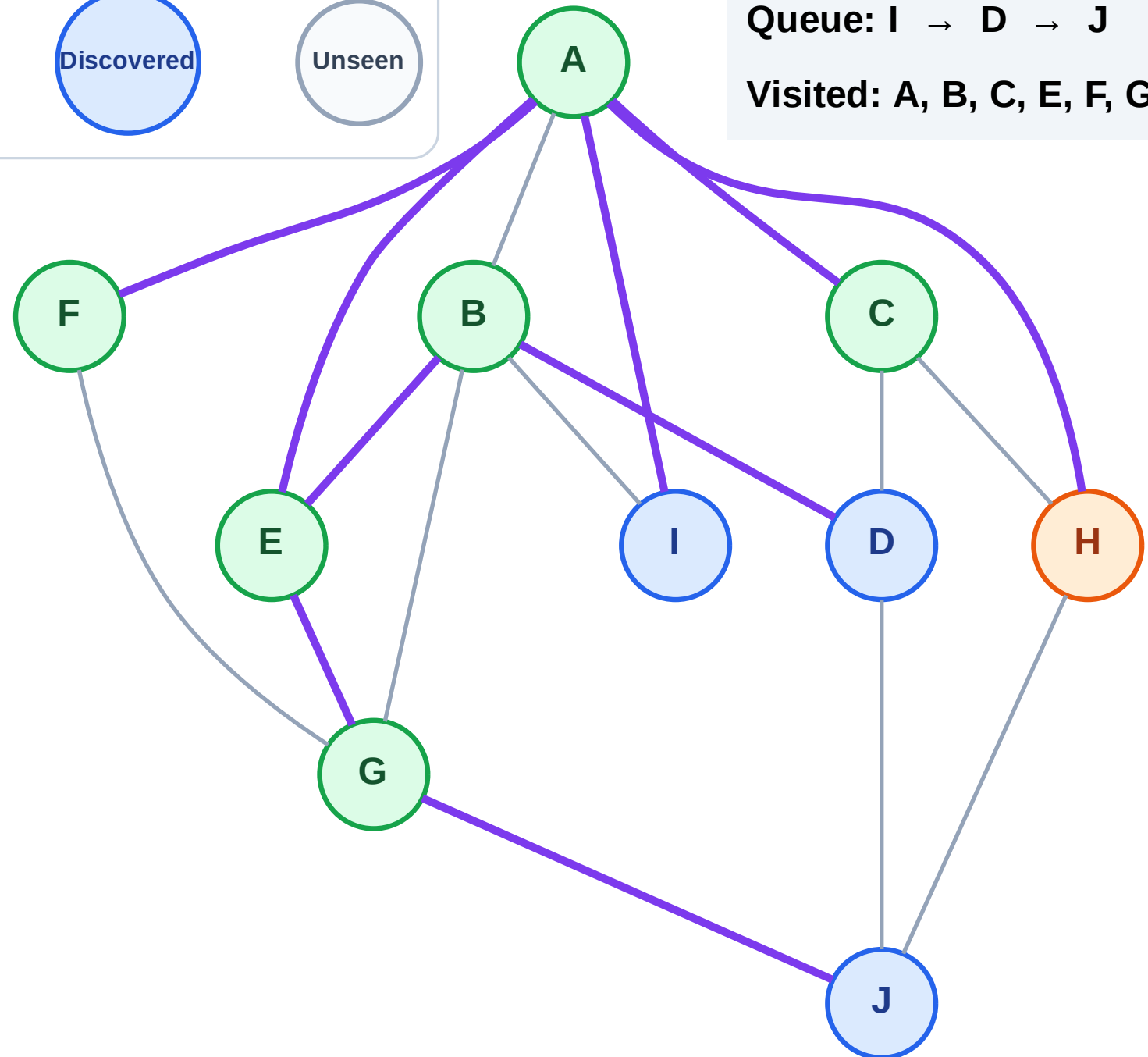
BFS Traversal

Step 14: Process node H

Legend



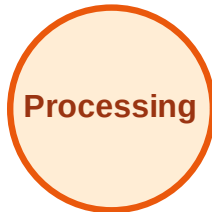
Queue: I → D → J
Visited: A, B, C, E, F, G



BFS Traversal

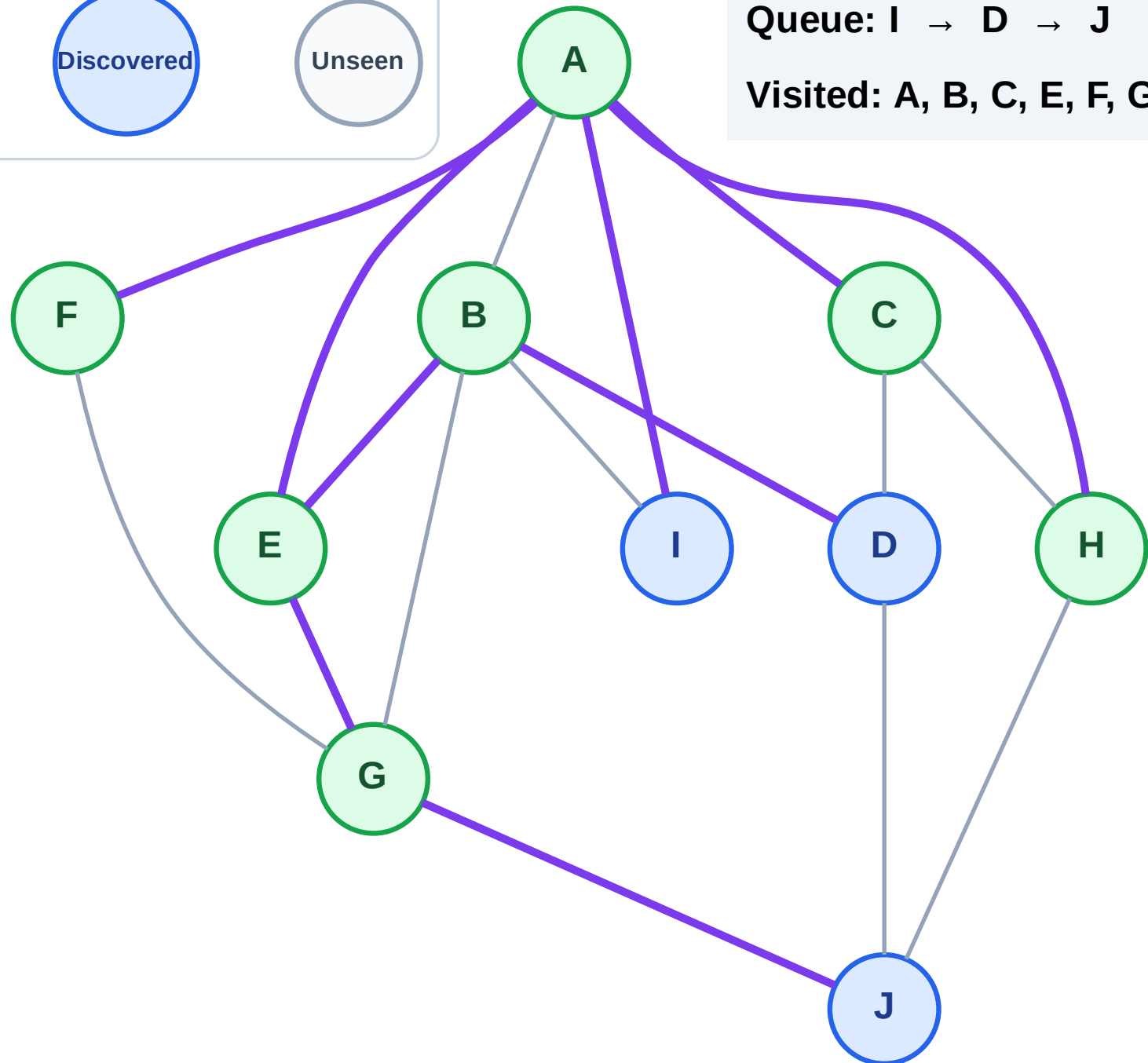
Step 15: No new neighbors from H

Legend



Queue: I → D → J

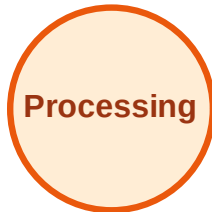
Visited: A, B, C, E, F, G, H



BFS Traversal

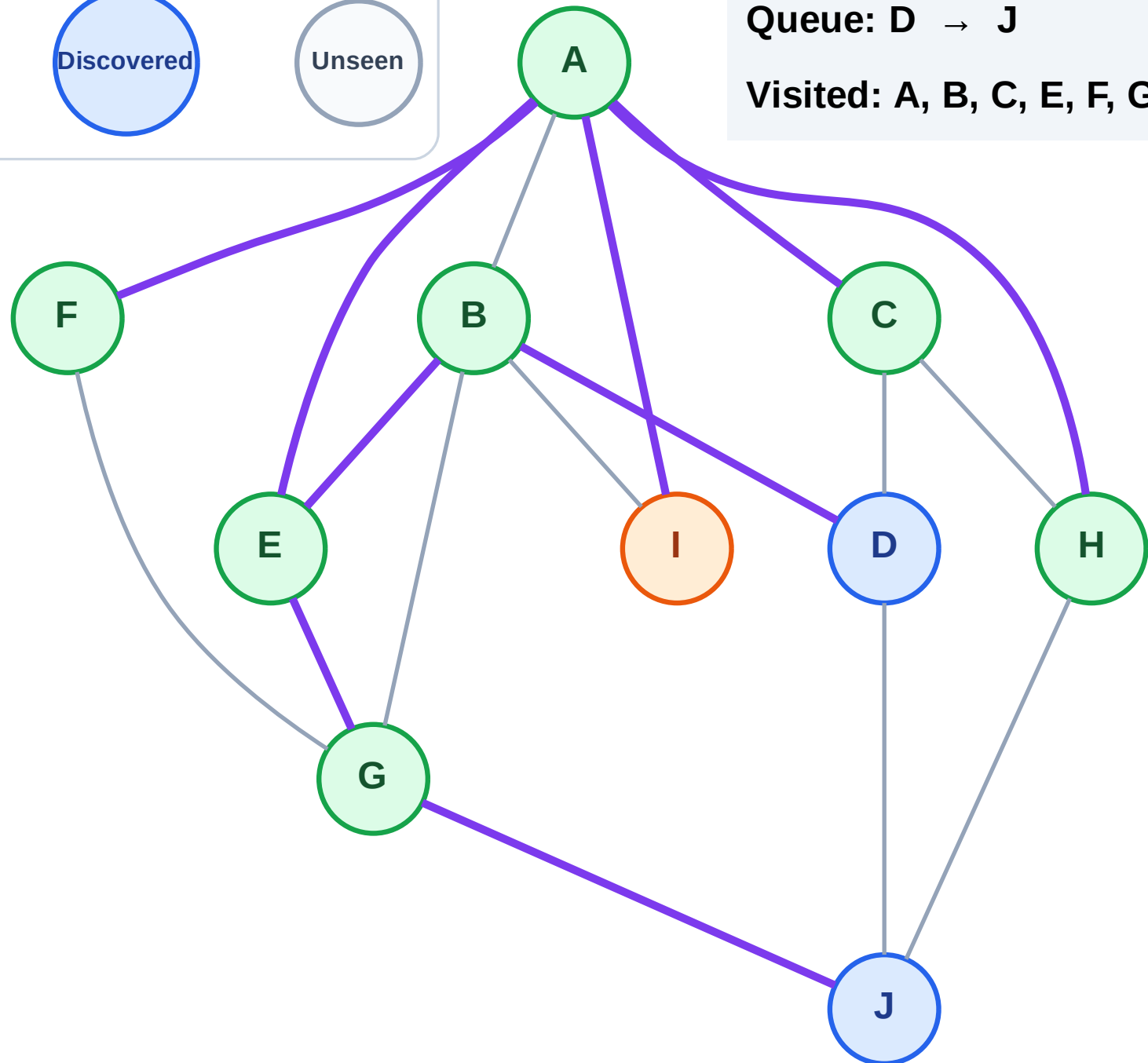
Step 16: Process node I

Legend



Queue: D → J

Visited: A, B, C, E, F, G, H



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

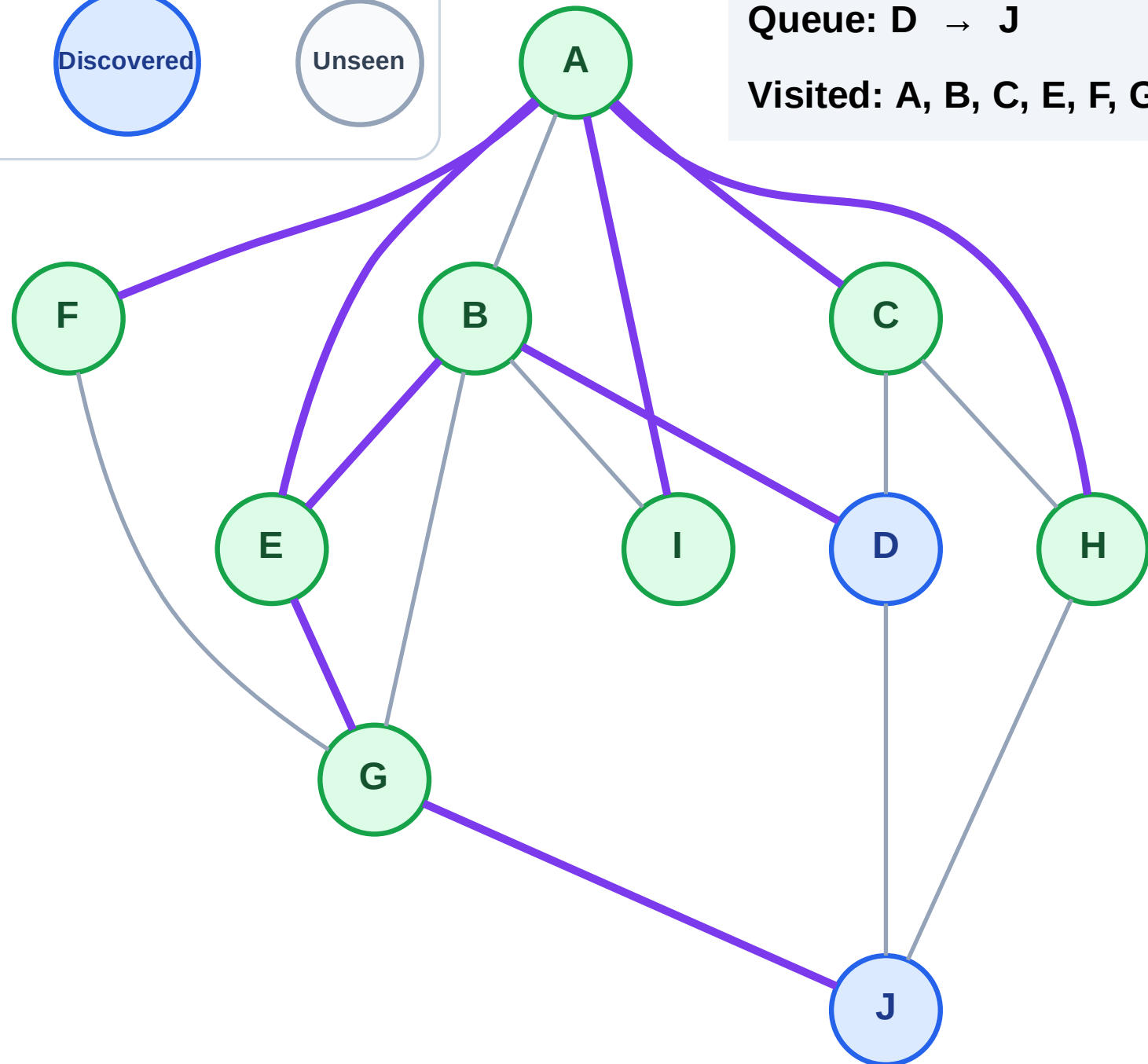
Processing

Discovered

Unseen

Queue: D → J

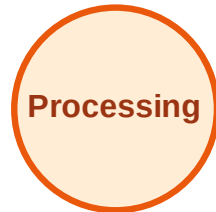
Visited: A, B, C, E, F, G, H, I



BFS Traversal

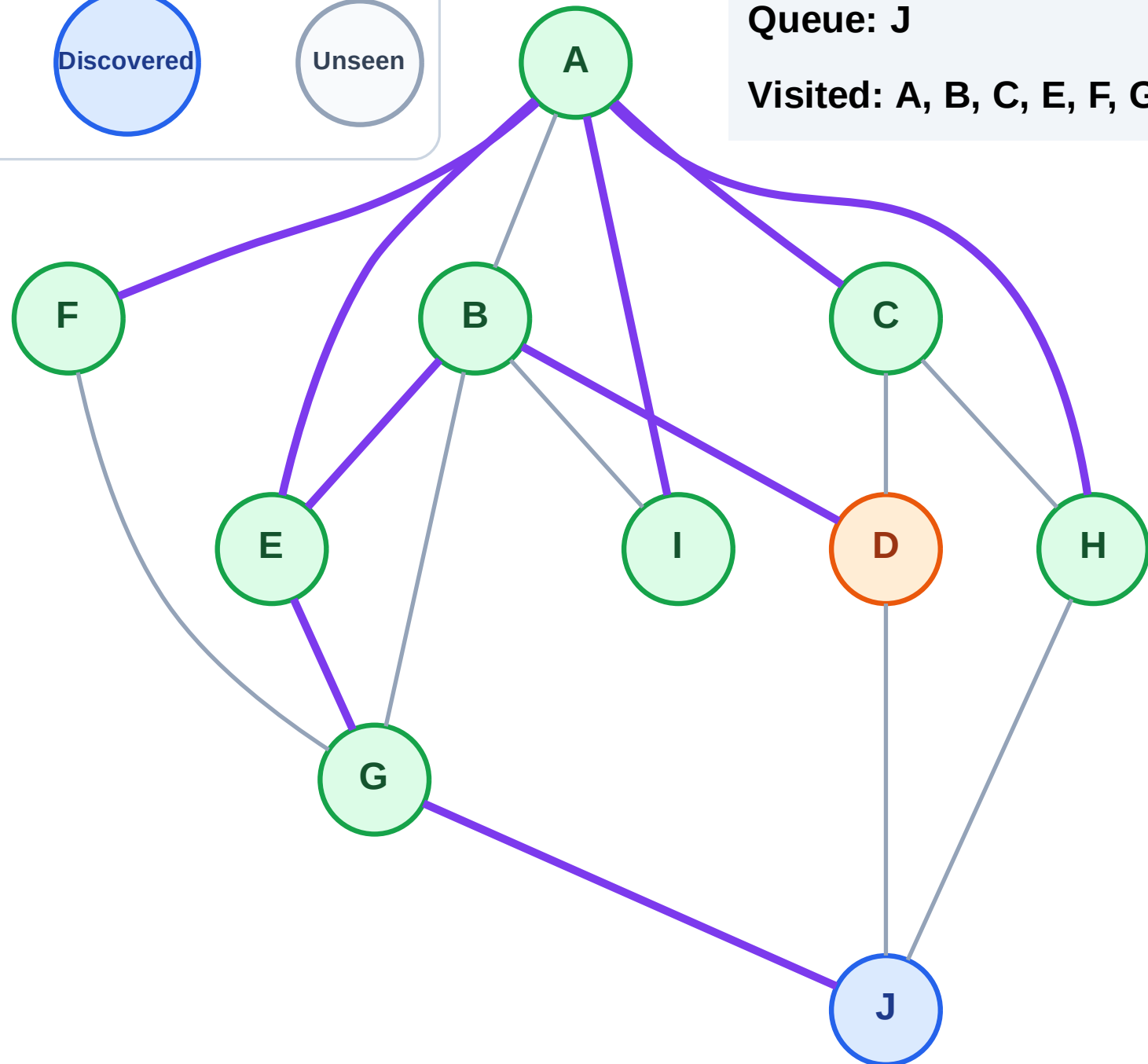
Step 18: Process node D

Legend



Queue: J

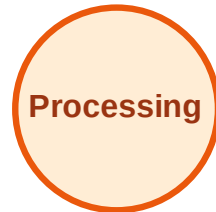
Visited: A, B, C, E, F, G, H, I



BFS Traversal

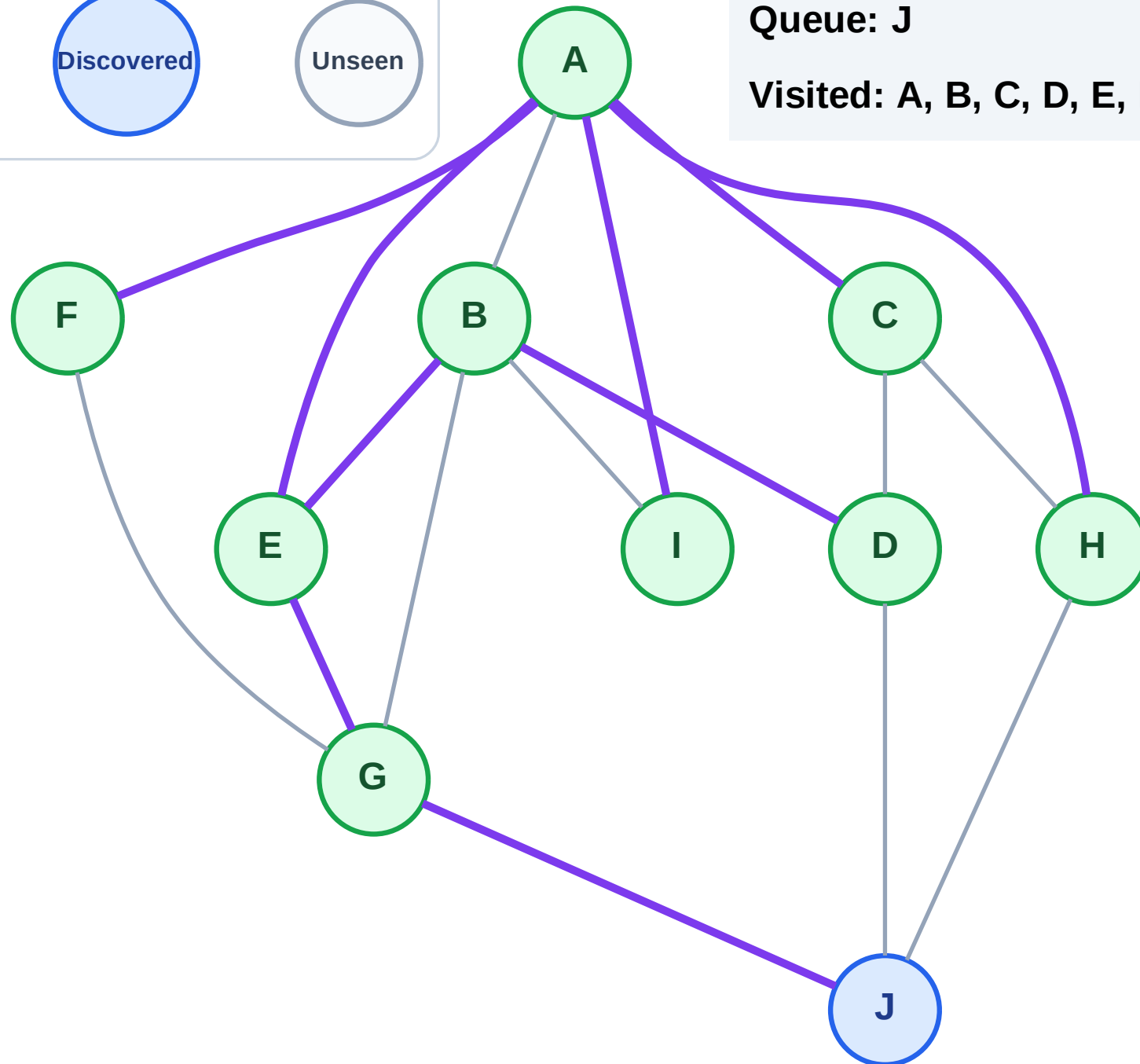
Step 19: No new neighbors from D

Legend



Queue: J

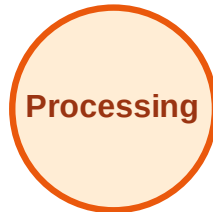
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

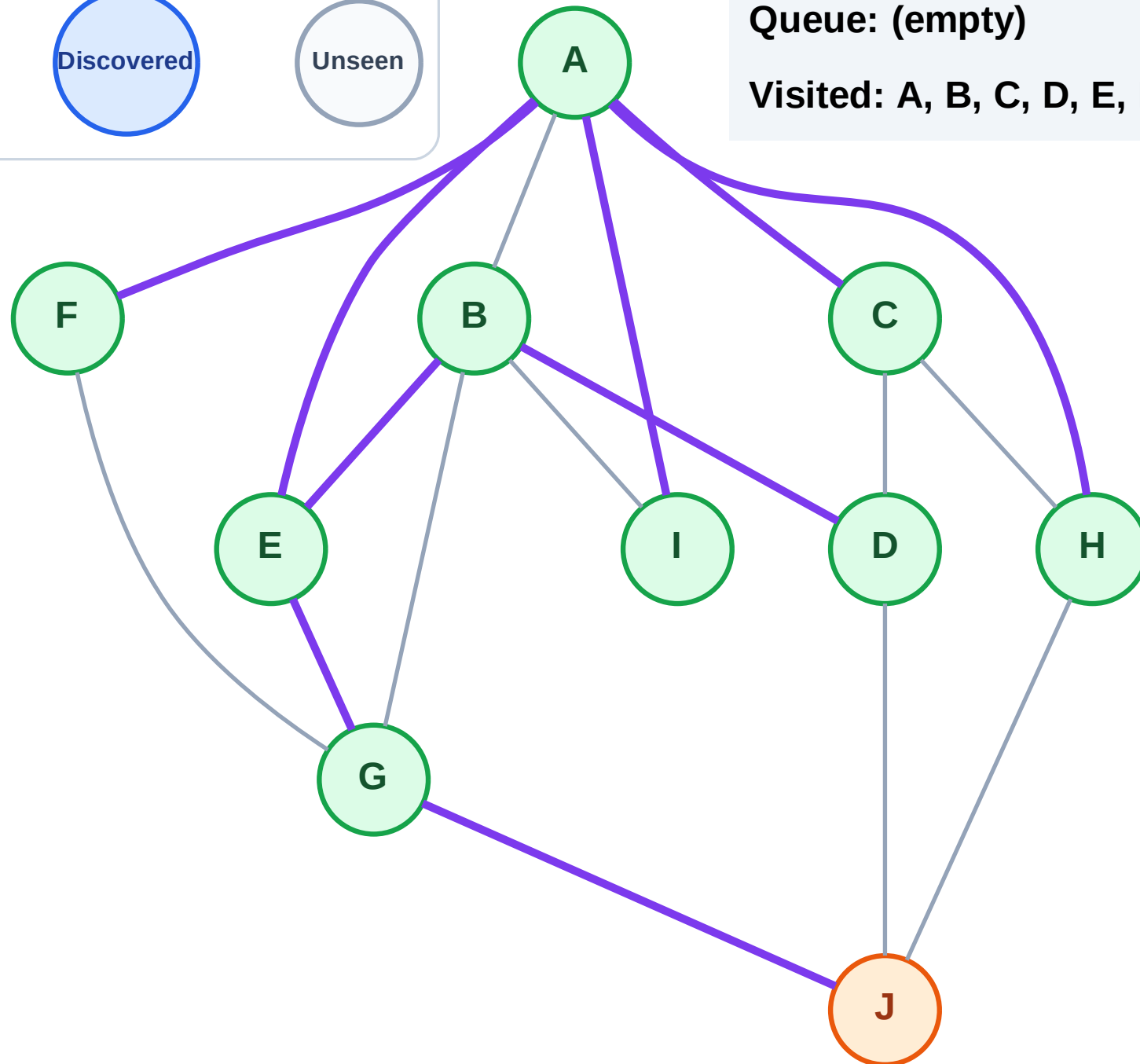
Step 20: Process node J

Legend



Queue: (empty)

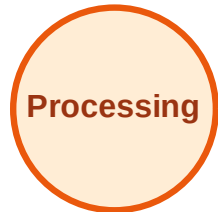
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

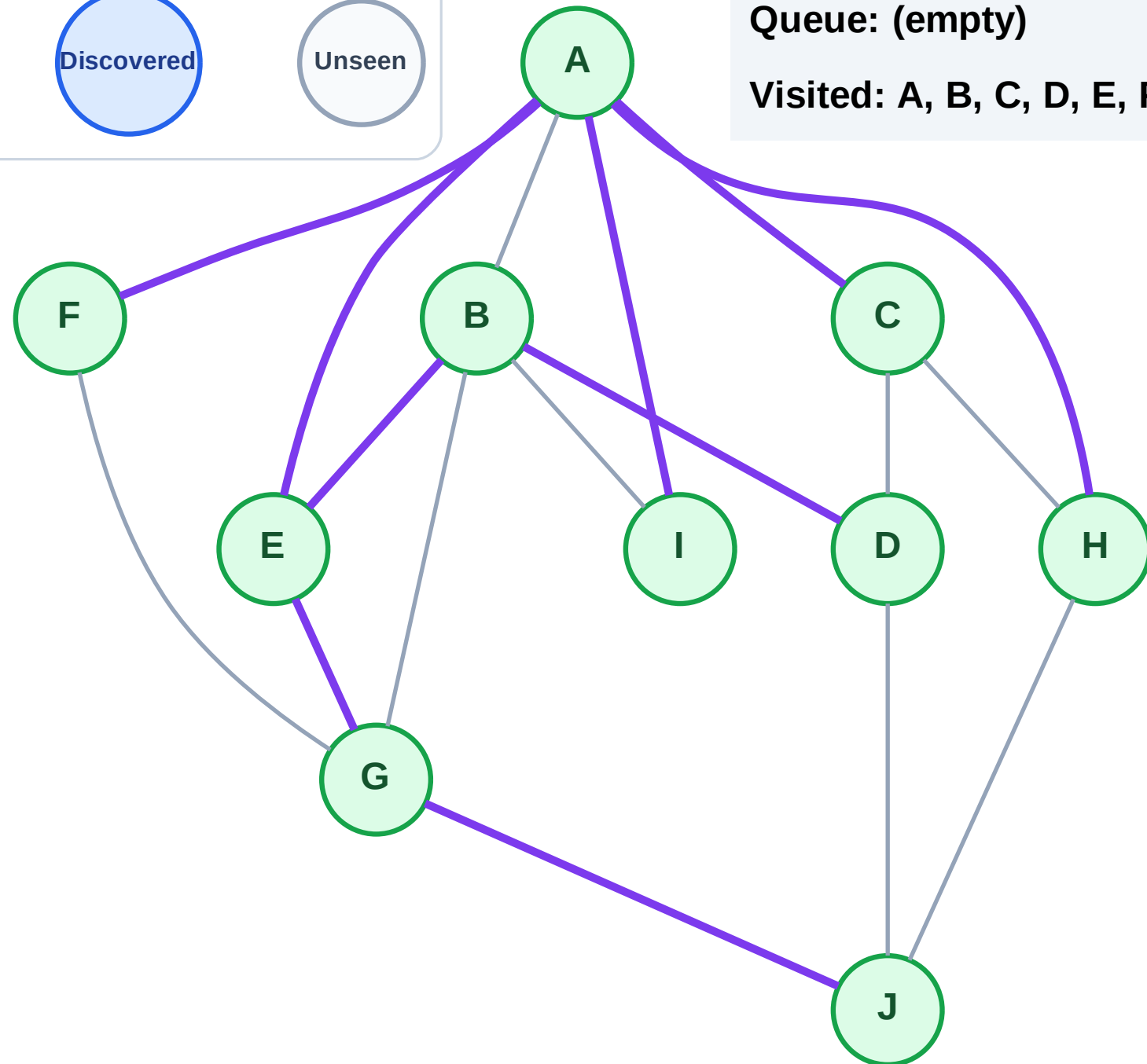
Step 21: No new neighbors from J

Legend



Queue: (empty)

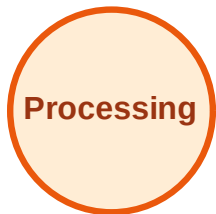
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

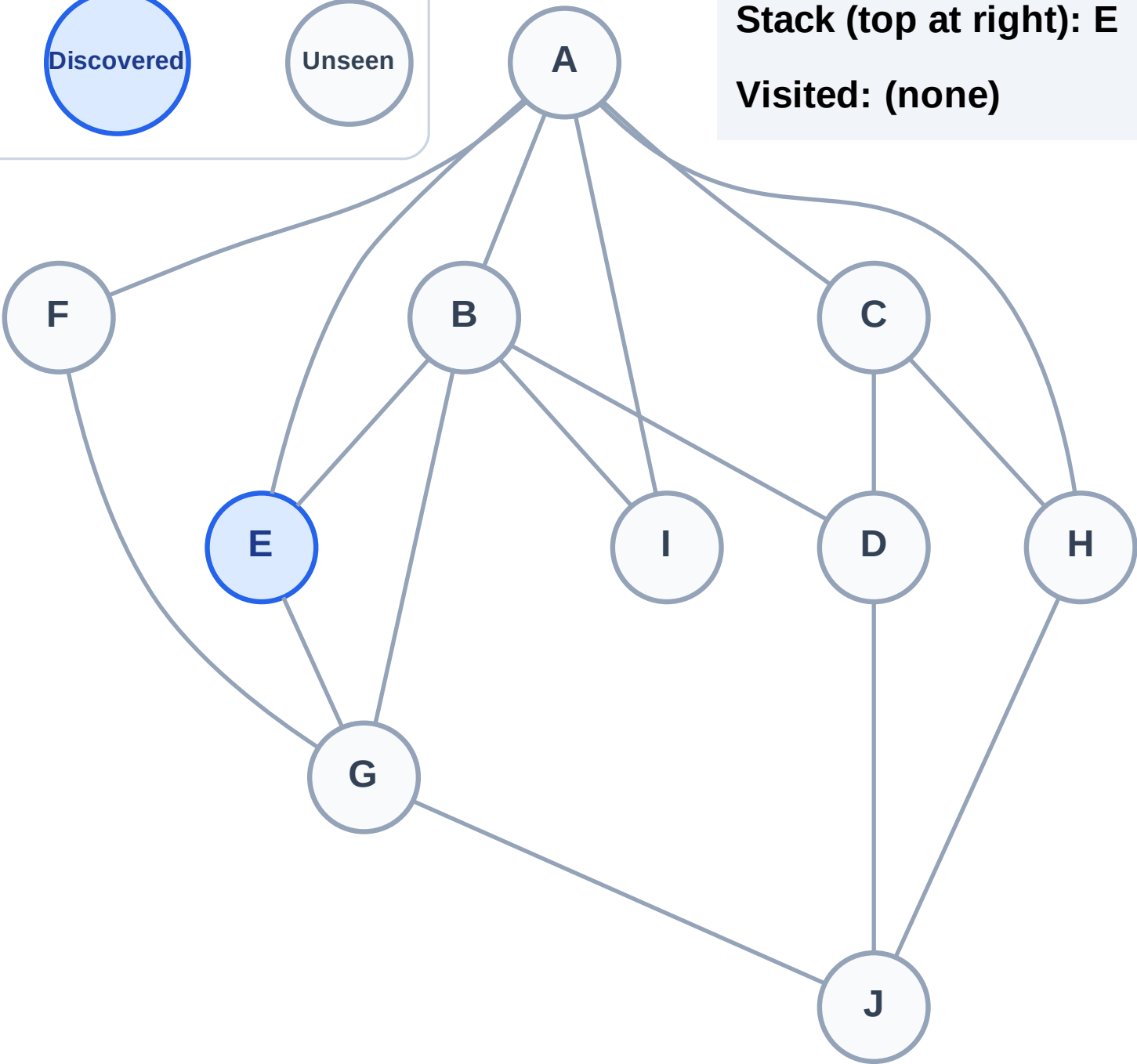
Step 1: Start at E

Legend



Stack (top at right): E

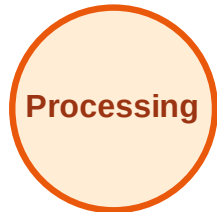
Visited: (none)



DFS Traversal

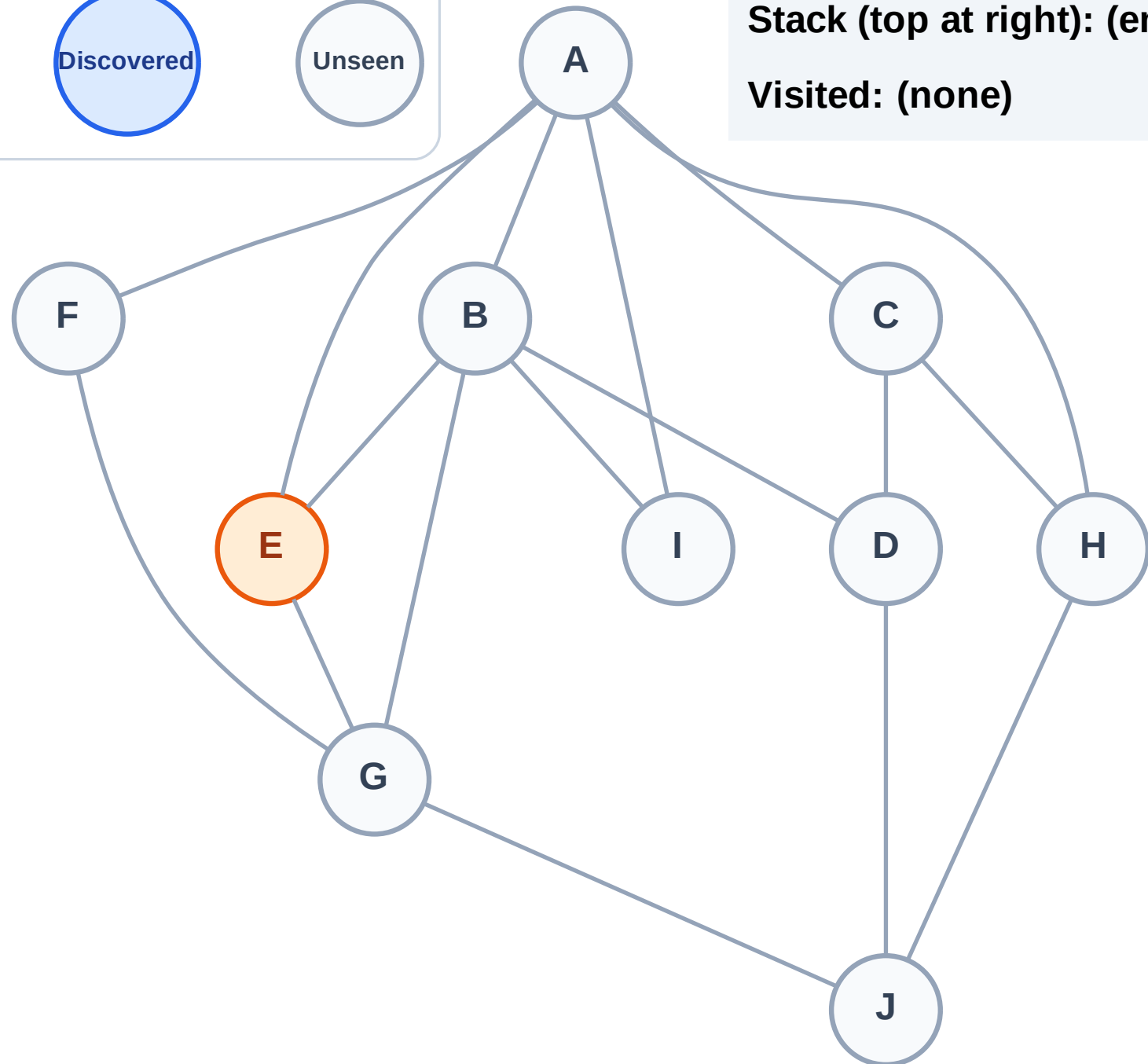
Step 2: Process node E

Legend



Stack (top at right): (empty)

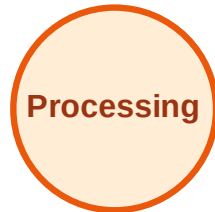
Visited: (none)



DFS Traversal

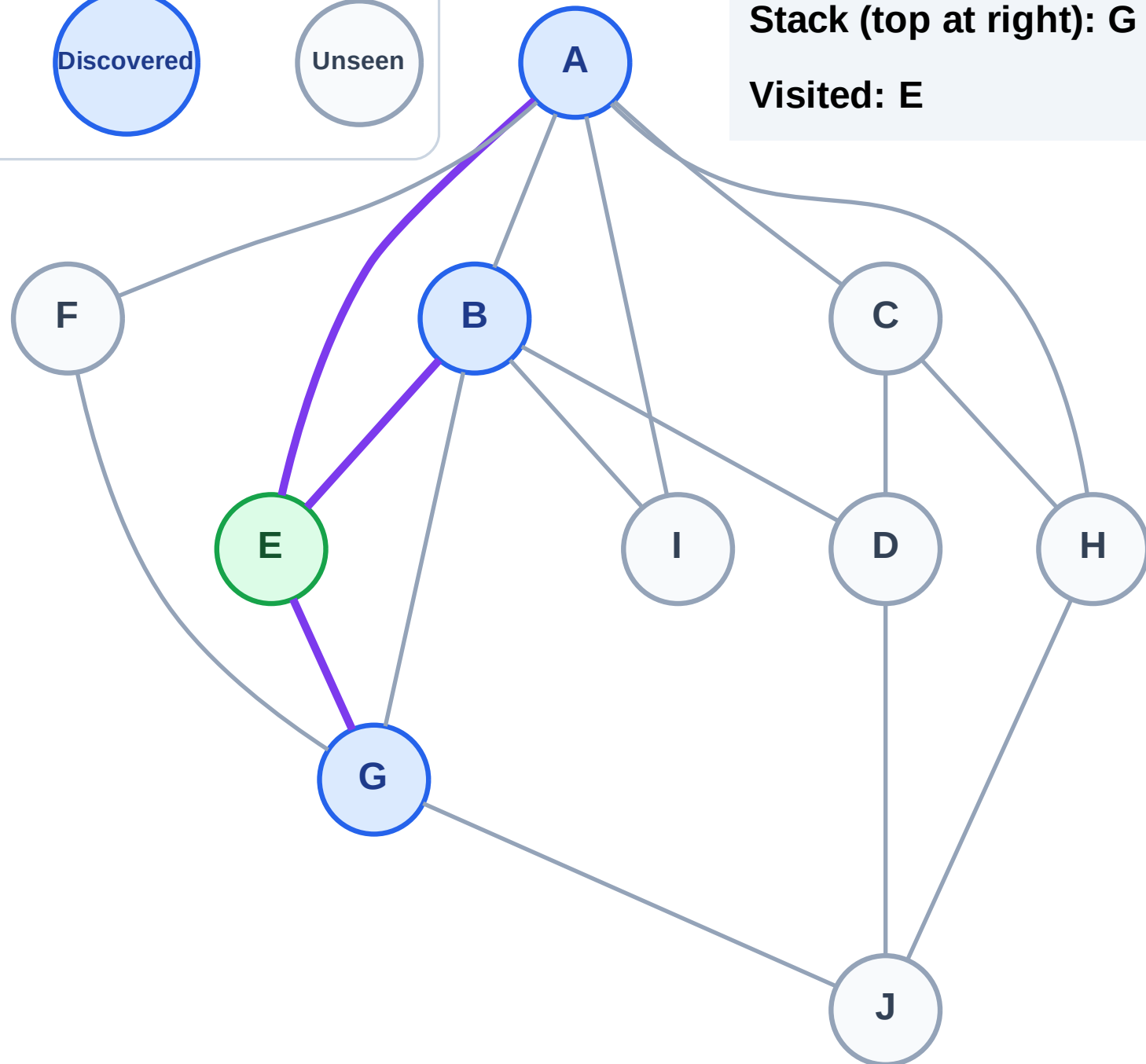
Step 3: Discover G, B, A from E

Legend



Stack (top at right): G | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

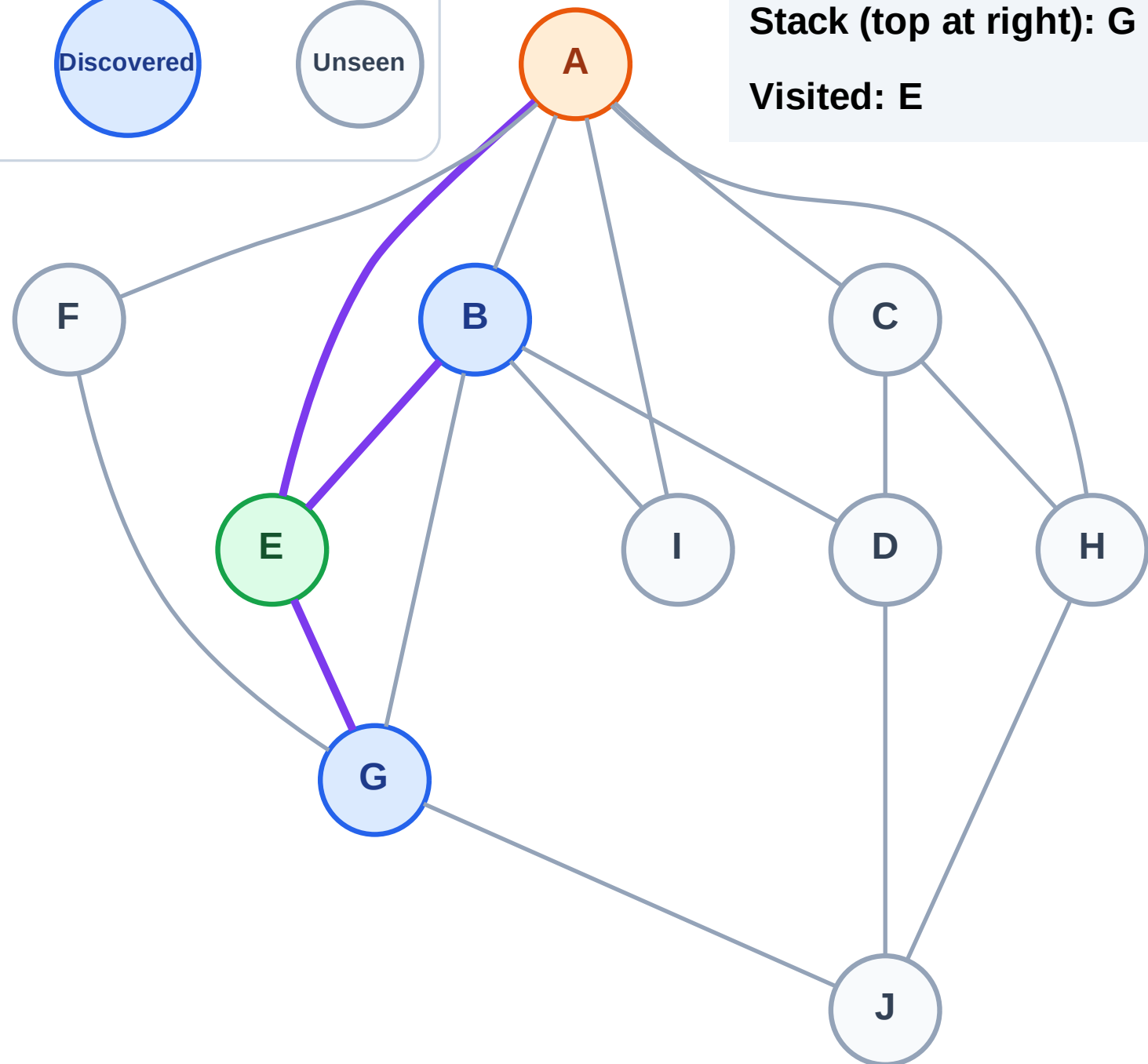
Processing

Discovered

Unseen

Stack (top at right): G | B

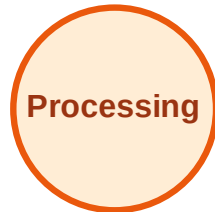
Visited: E



DFS Traversal

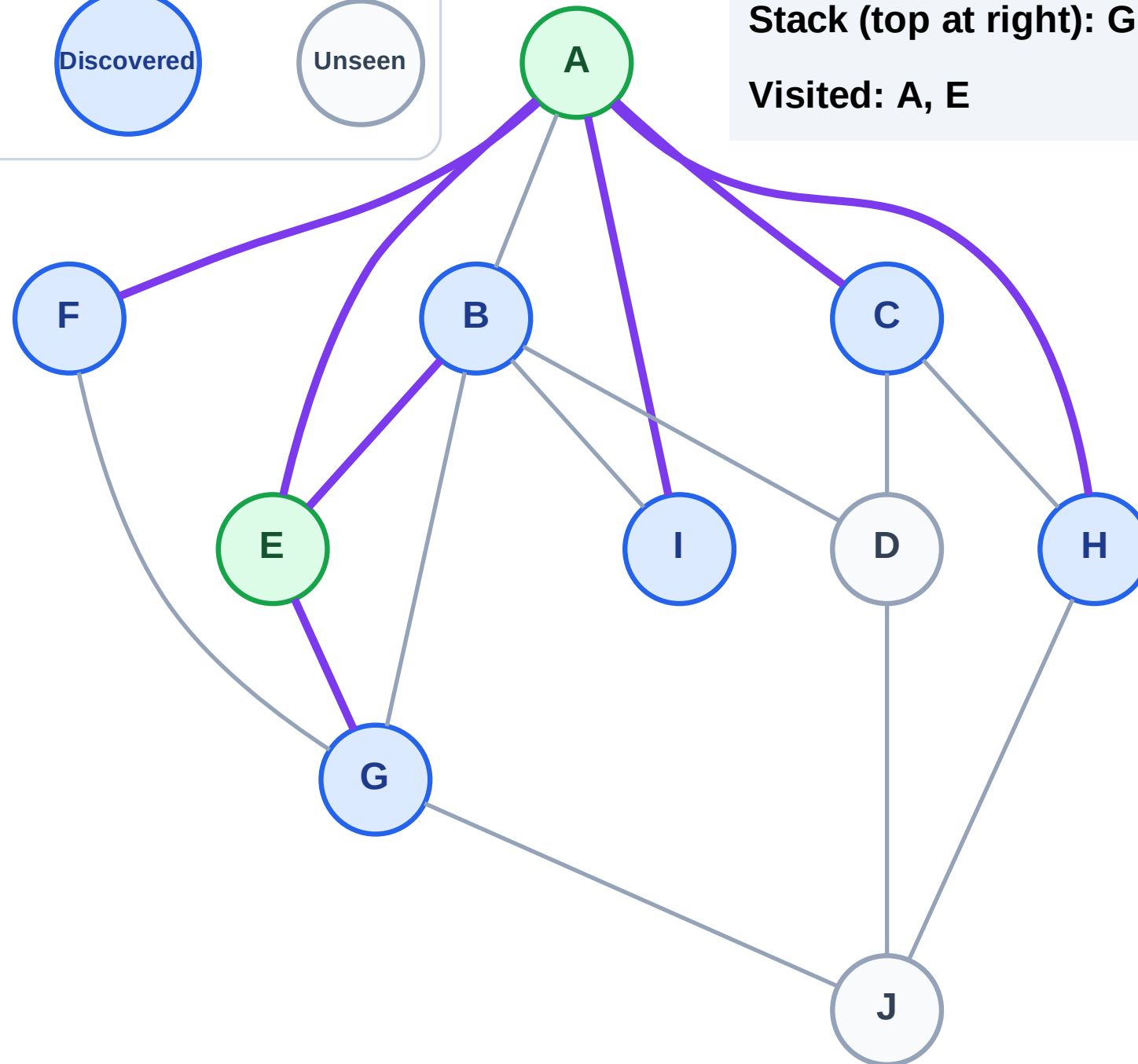
Step 5: Discover I, H, F, C from A

Legend



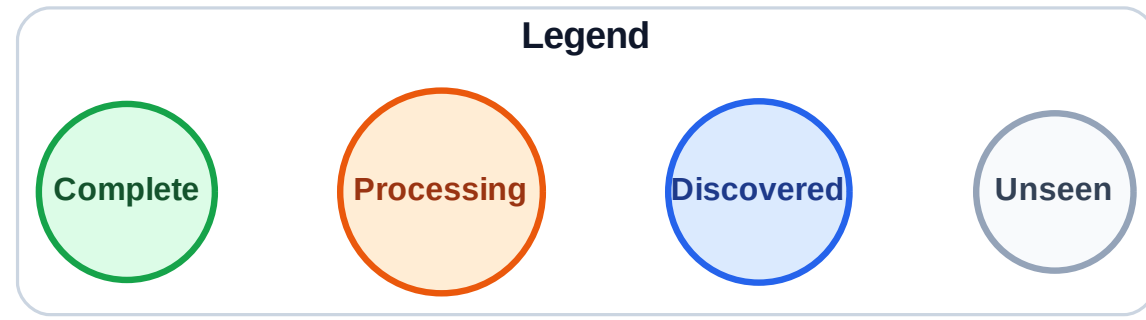
Stack (top at right): G | B | I | H | F | C

Visited: A, E

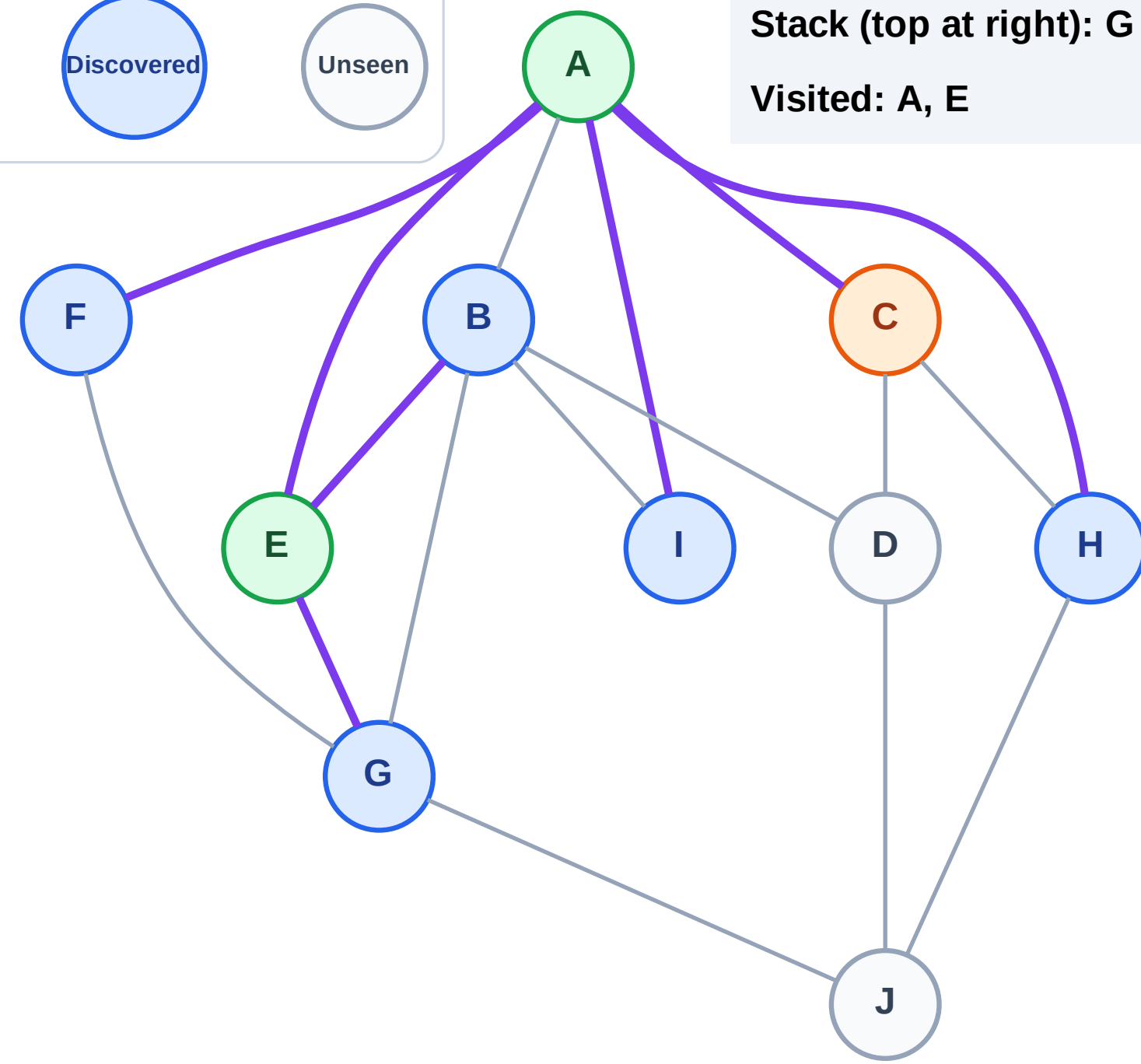


DFS Traversal

Step 6: Process node C



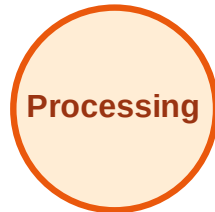
Stack (top at right): G | B | I | H | F
Visited: A, E



DFS Traversal

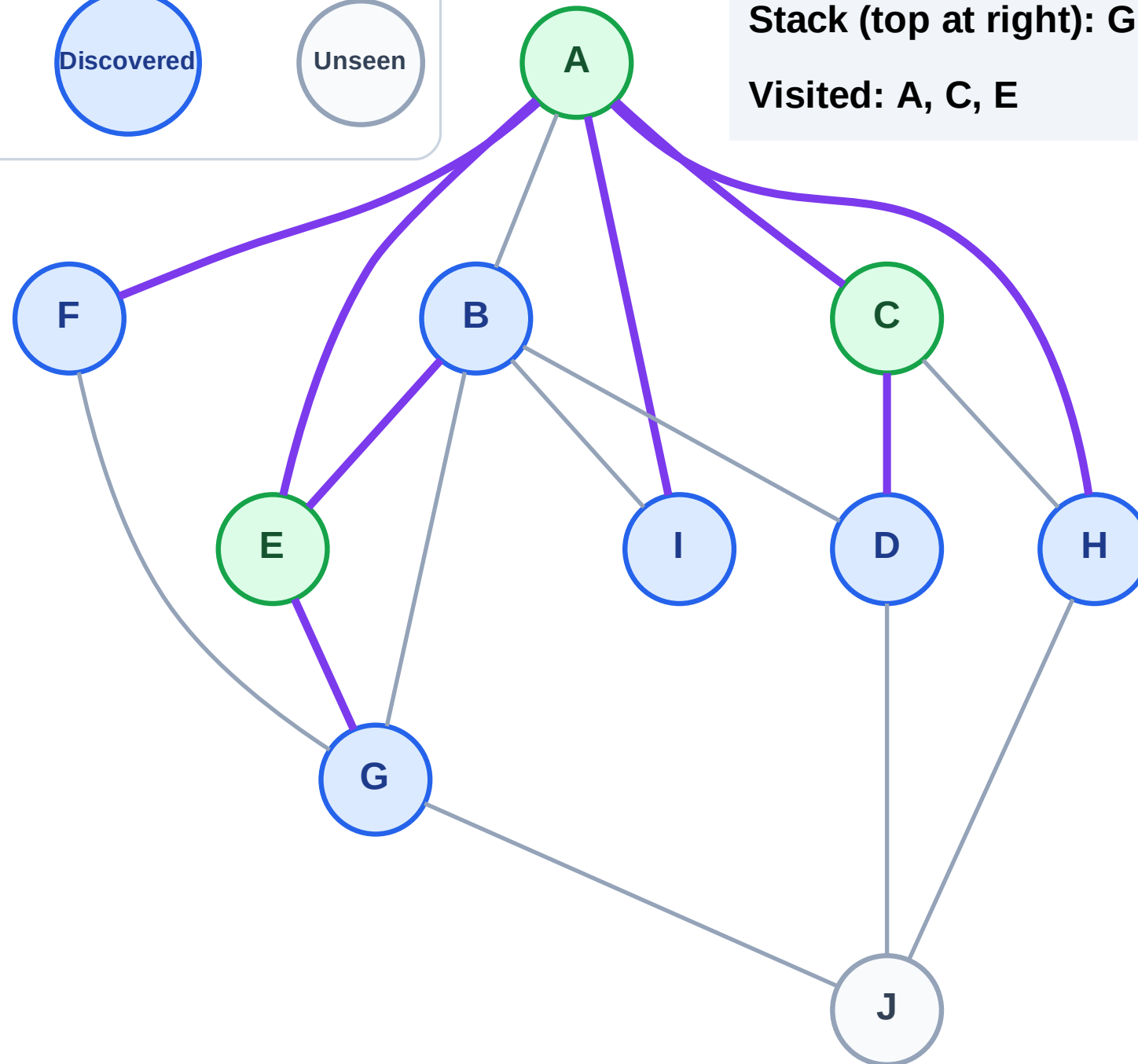
Step 7: Discover D from C

Legend



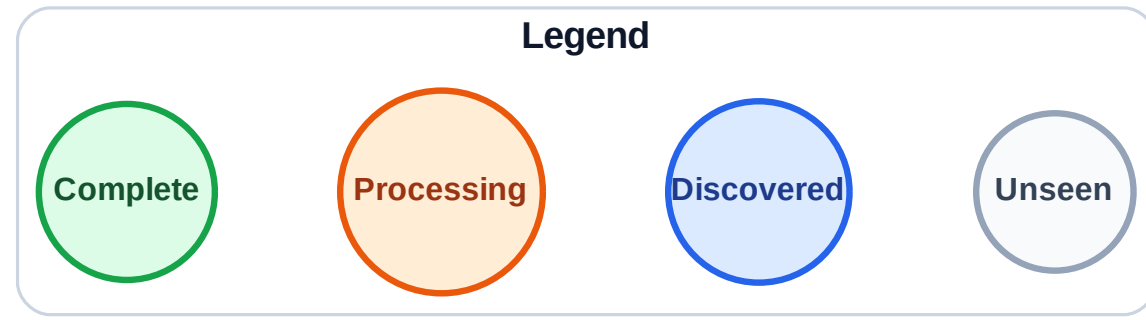
Stack (top at right): G | B | I | H | F | D

Visited: A, C, E

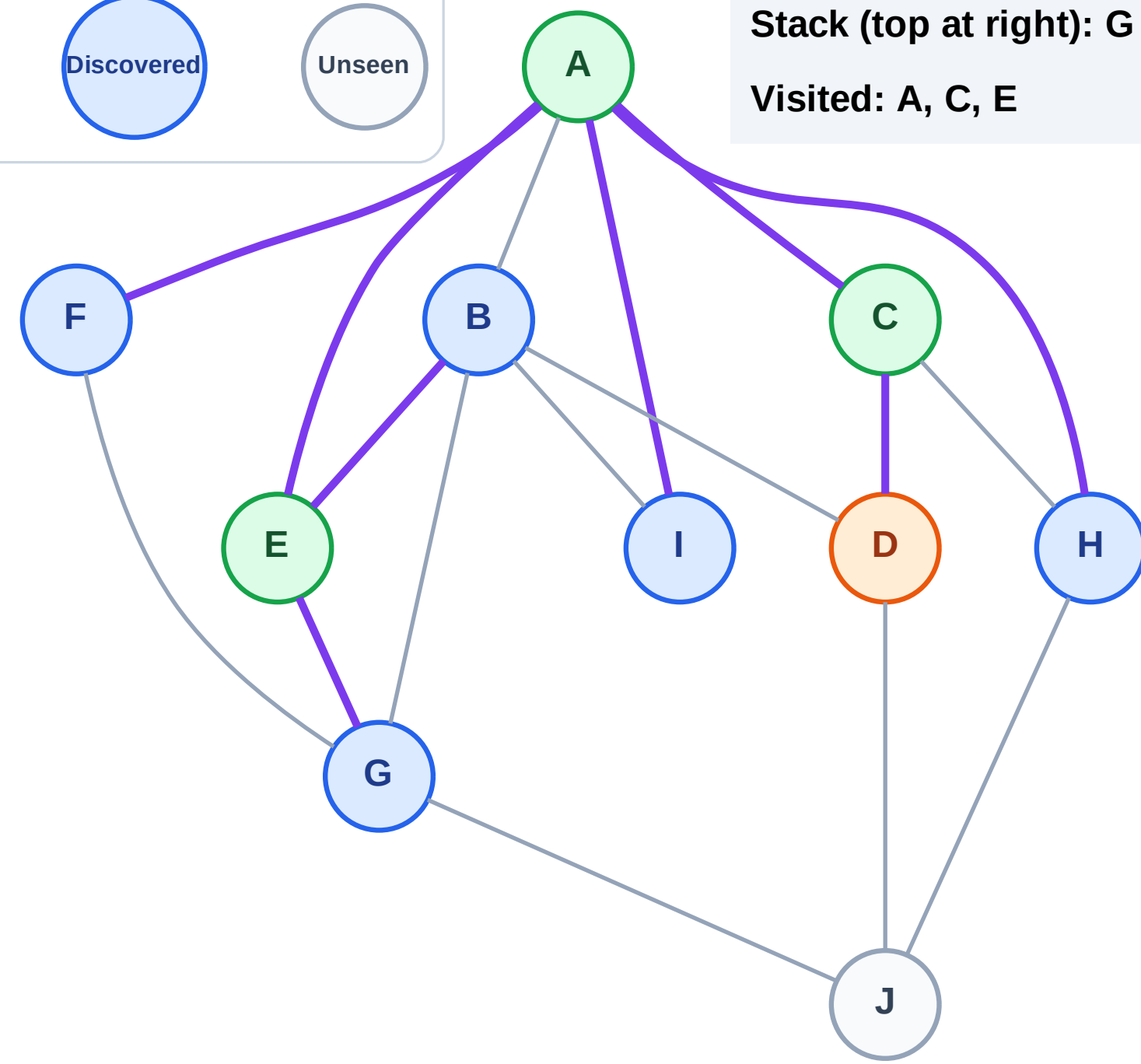


DFS Traversal

Step 8: Process node D



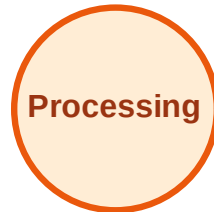
Stack (top at right): G | B | I | H | F
Visited: A, C, E



DFS Traversal

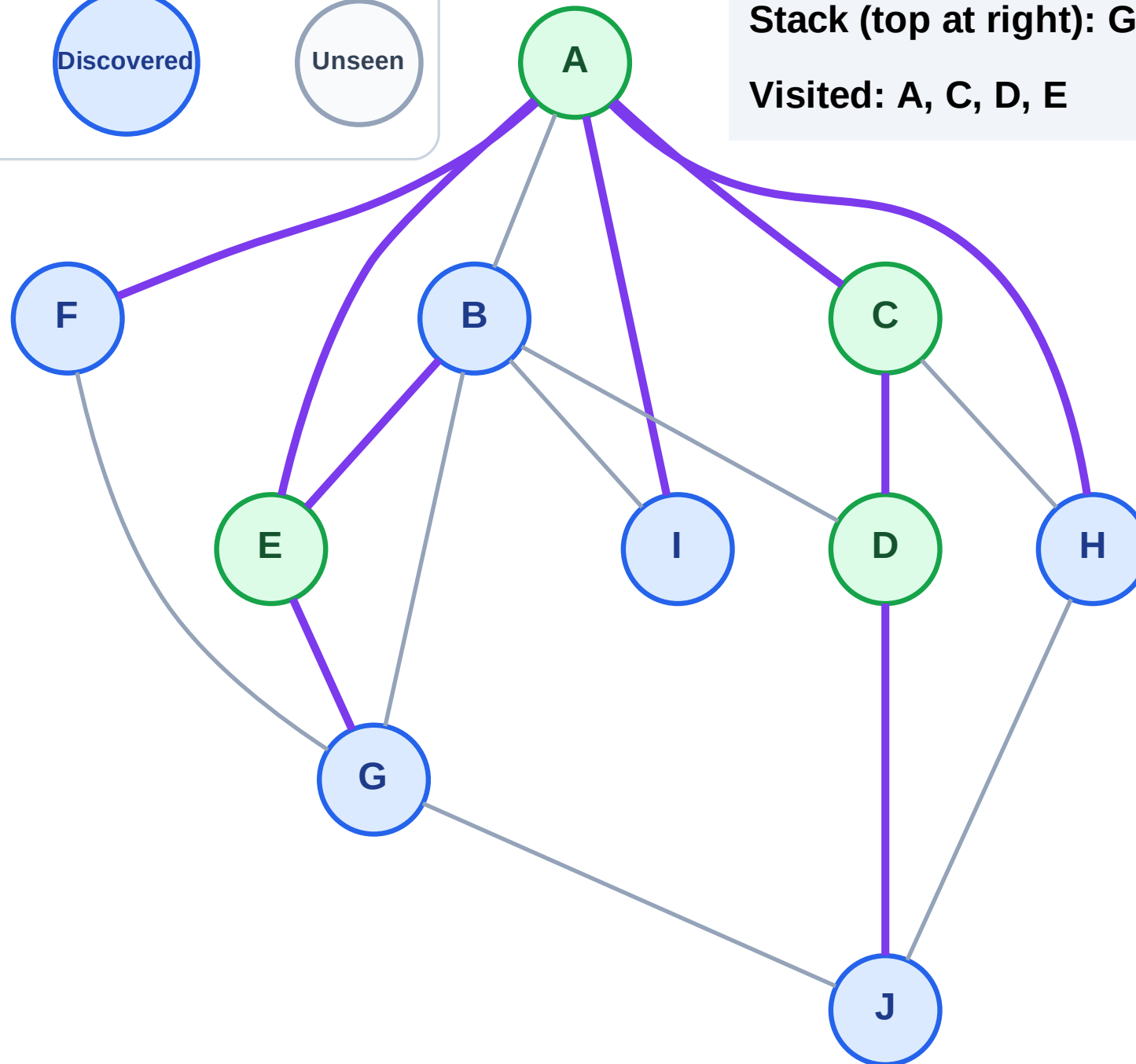
Step 9: Discover J from D

Legend



Stack (top at right): G | B | I | H | F | J

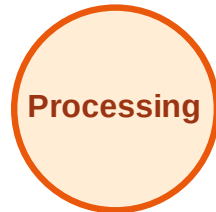
Visited: A, C, D, E



DFS Traversal

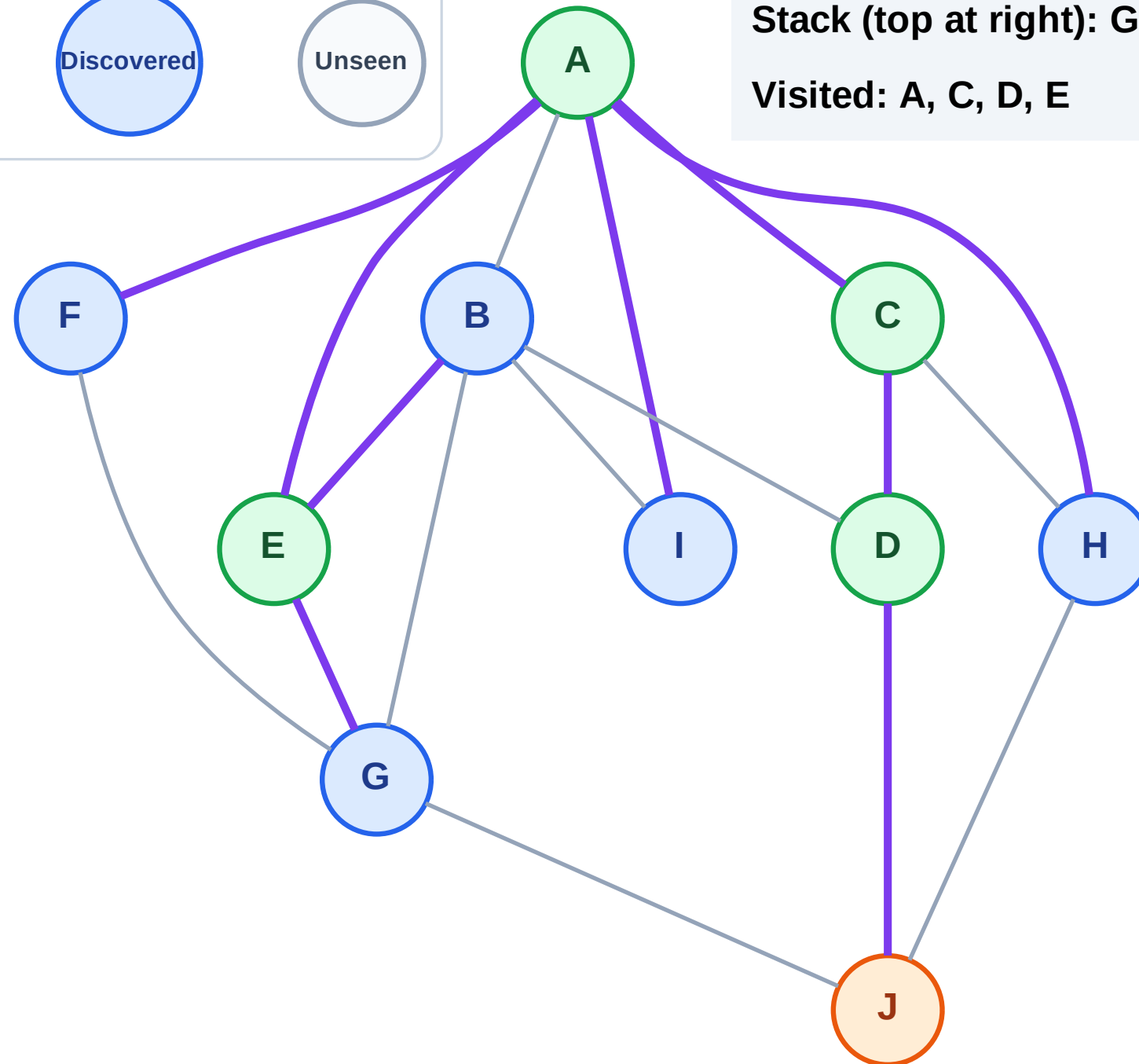
Step 10: Process node J

Legend



Stack (top at right): G | B | I | H | F

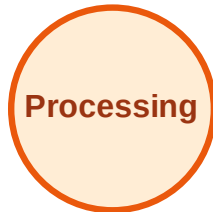
Visited: A, C, D, E



DFS Traversal

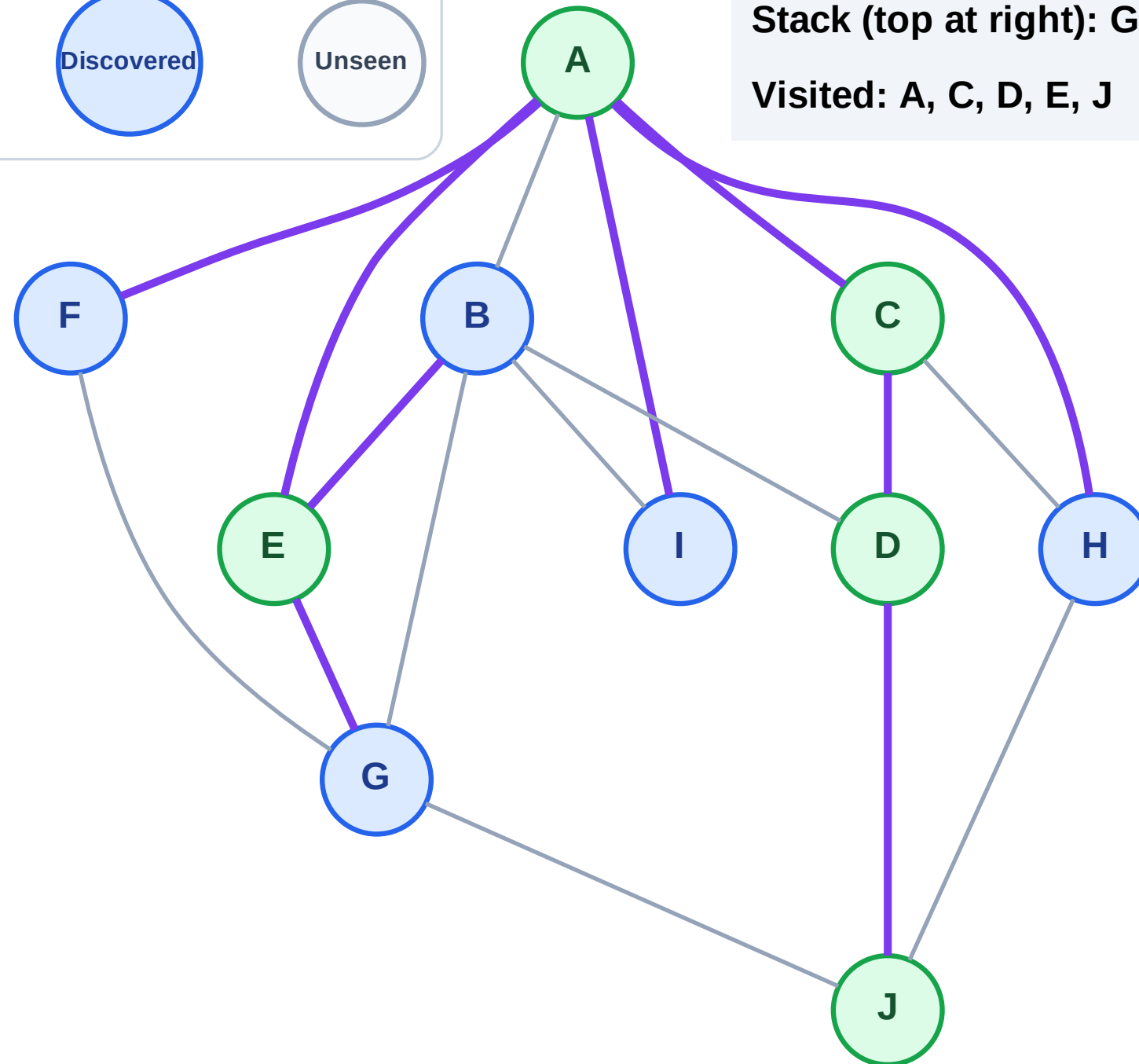
Step 11: No new neighbors from J

Legend



Stack (top at right): G | B | I | H | F

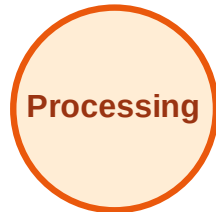
Visited: A, C, D, E, J



DFS Traversal

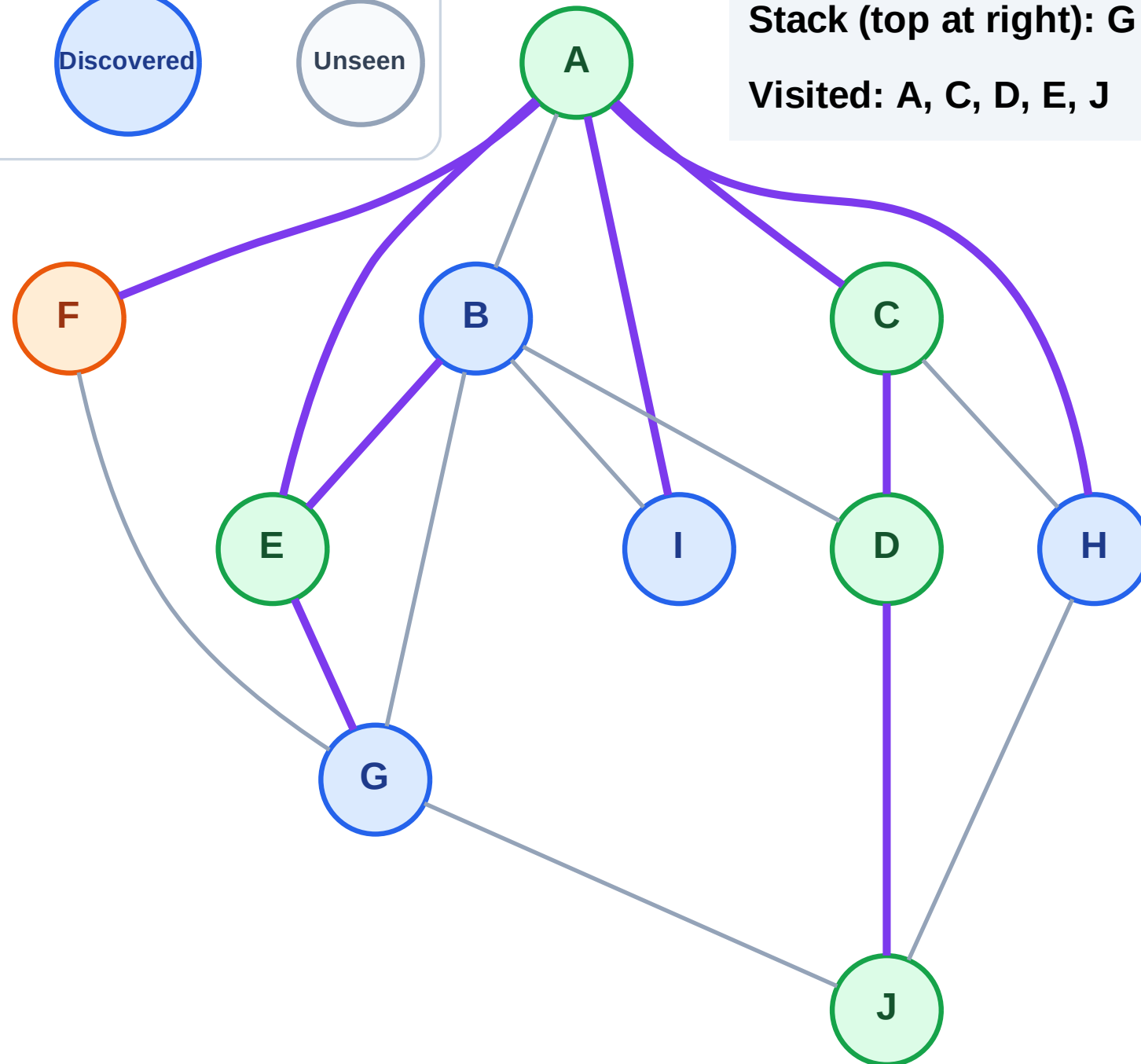
Step 12: Process node F

Legend



Stack (top at right): G | B | I | H

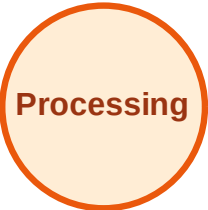
Visited: A, C, D, E, J



DFS Traversal

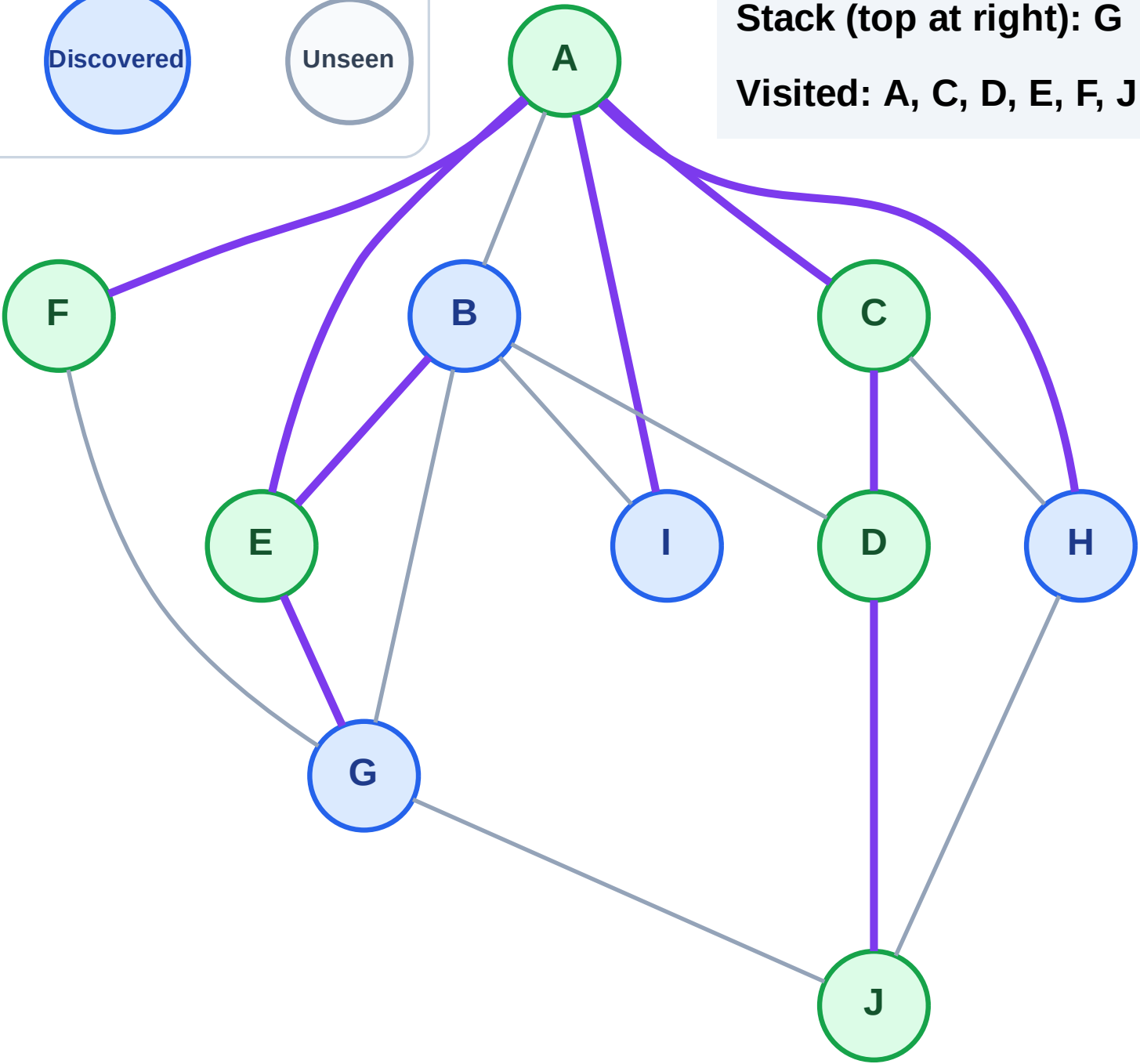
Step 13: No new neighbors from F

Legend



Stack (top at right): G | B | I | H

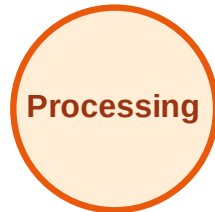
Visited: A, C, D, E, F, J



DFS Traversal

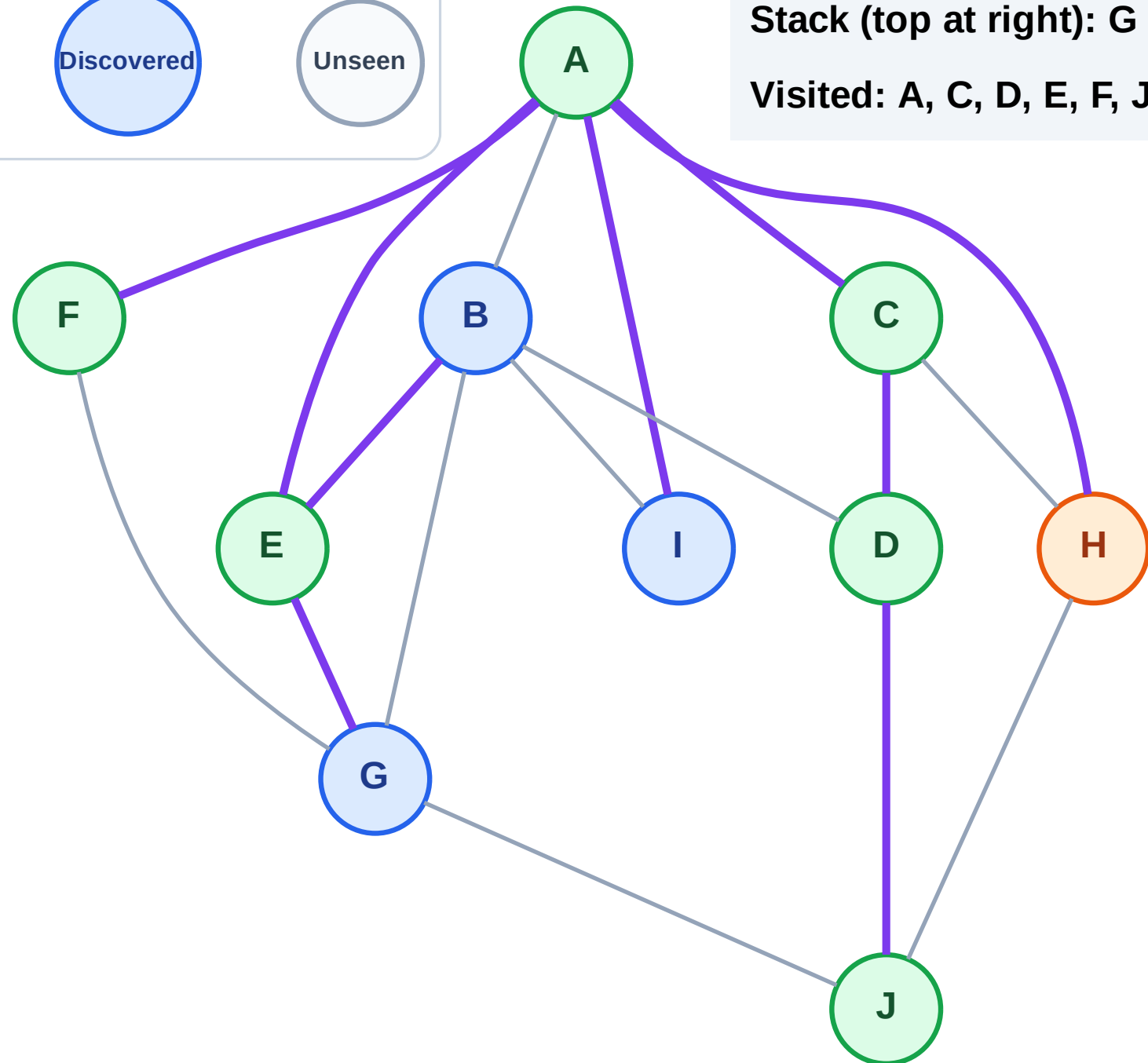
Step 14: Process node H

Legend



Stack (top at right): G | B | I

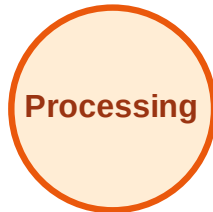
Visited: A, C, D, E, F, J



DFS Traversal

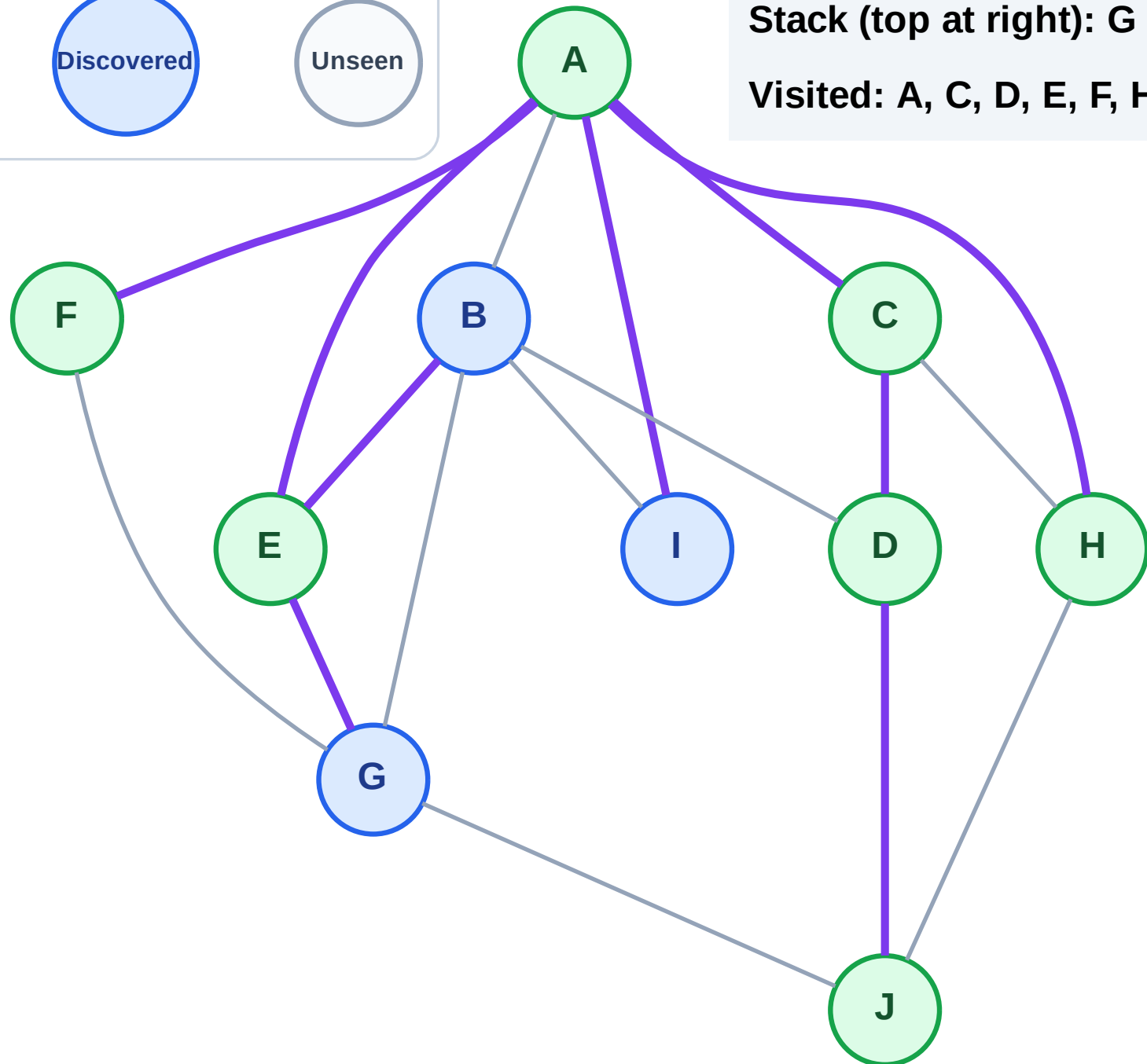
Step 15: No new neighbors from H

Legend



Stack (top at right): G | B | I

Visited: A, C, D, E, F, H, J



DFS Traversal

Step 16: Process node I

Legend

Complete

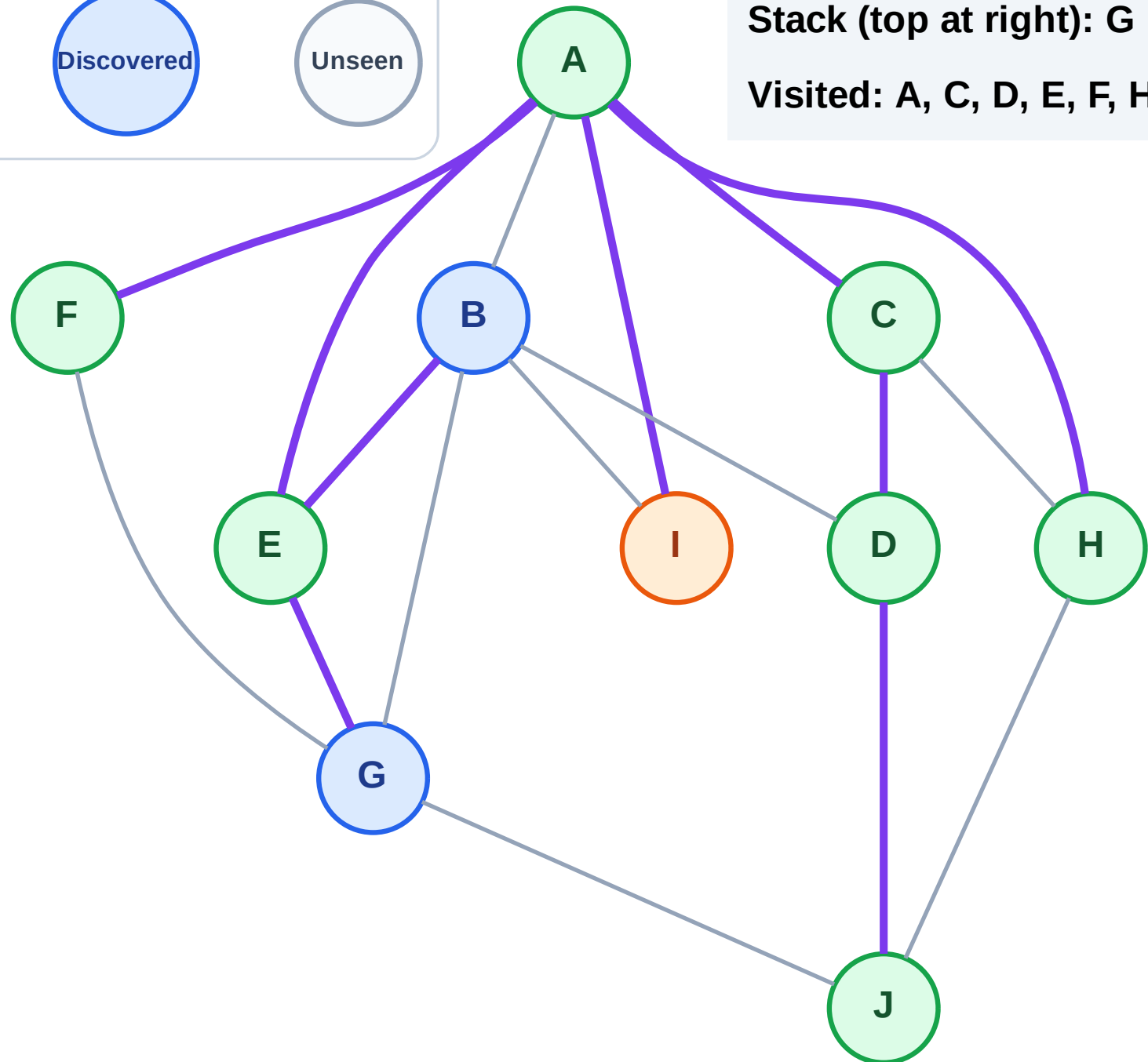
Processing

Discovered

Unseen

Stack (top at right): G | B

Visited: A, C, D, E, F, H, J



DFS Traversal

Step 17: No new neighbors from I

Legend

Complete

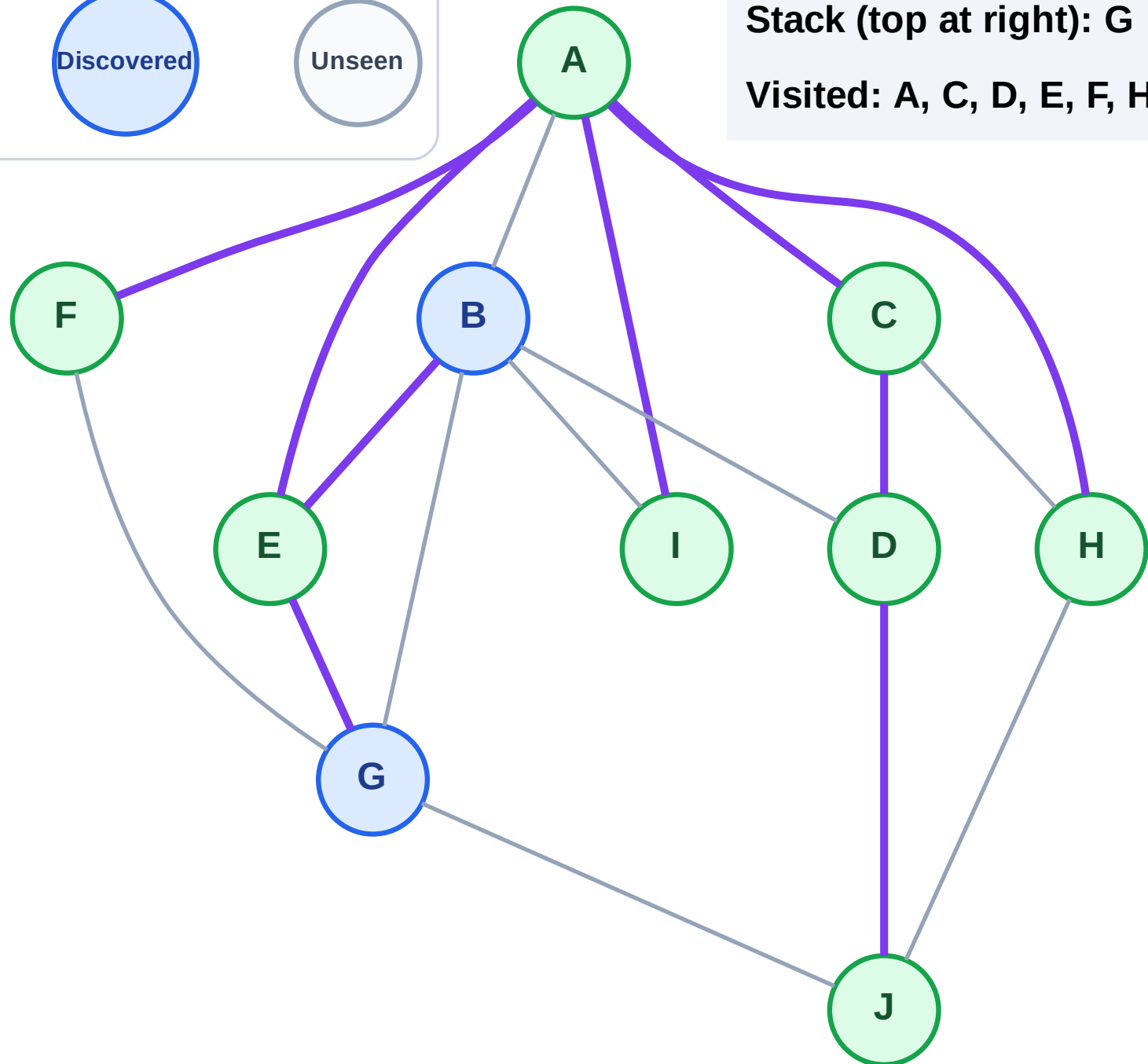
Processing

Discovered

Unseen

Stack (top at right): G | B

Visited: A, C, D, E, F, H, I, J



DFS Traversal

Step 18: Process node B

Legend

Complete

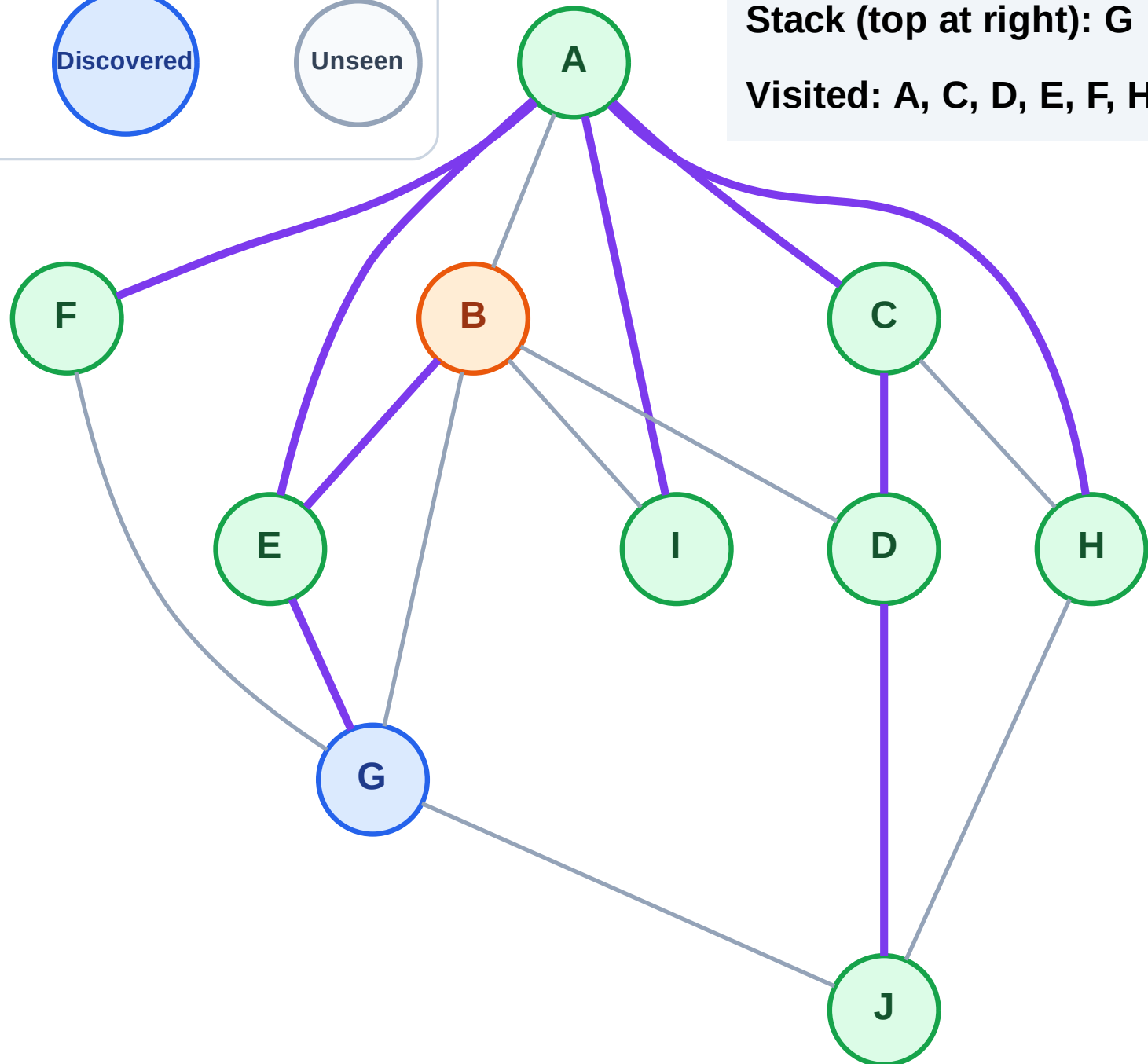
Processing

Discovered

Unseen

Stack (top at right): G

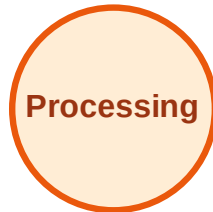
Visited: A, C, D, E, F, H, I, J



DFS Traversal

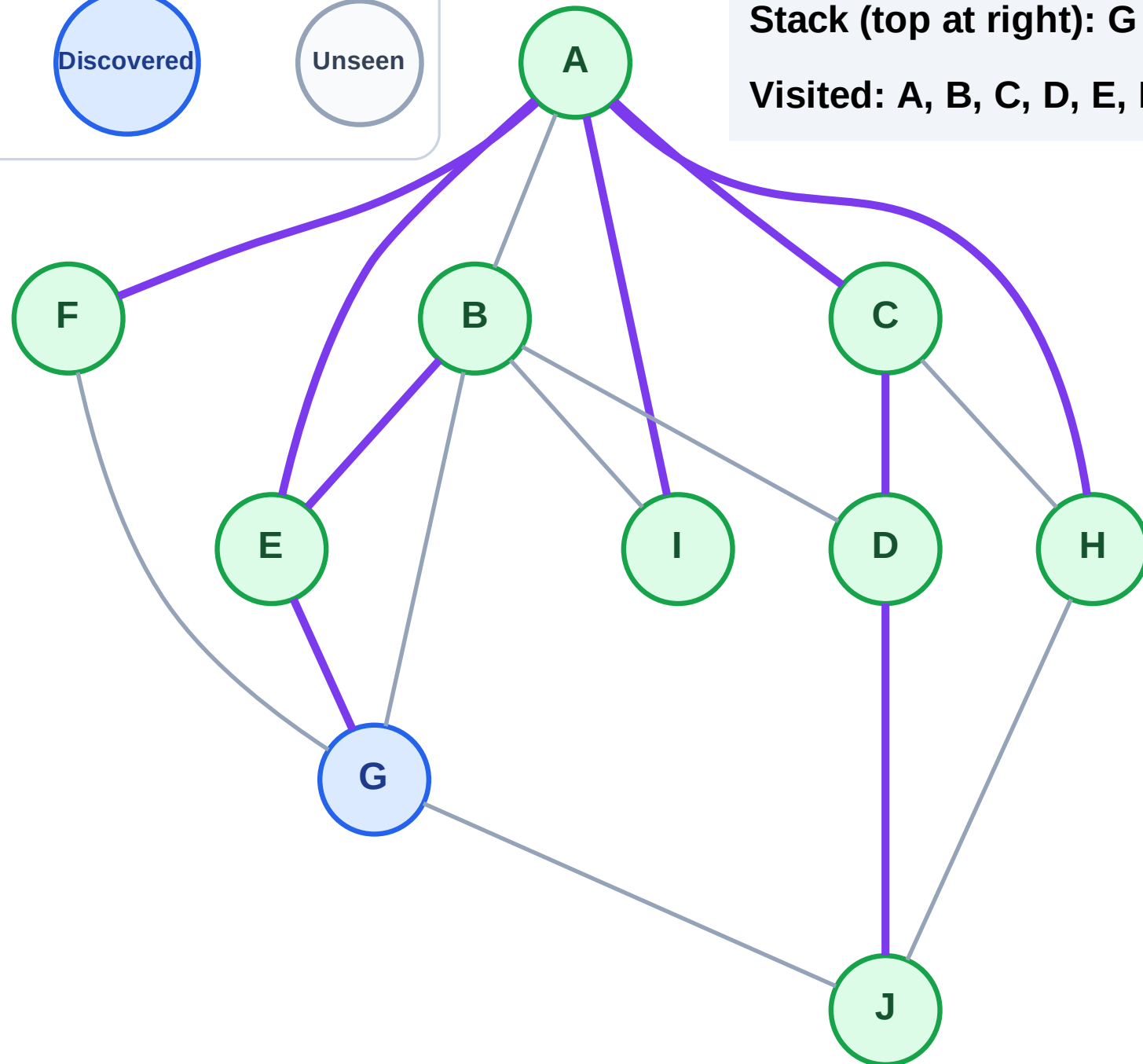
Step 19: No new neighbors from B

Legend



Stack (top at right): G

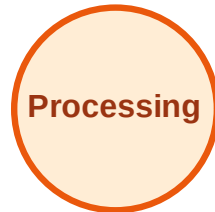
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

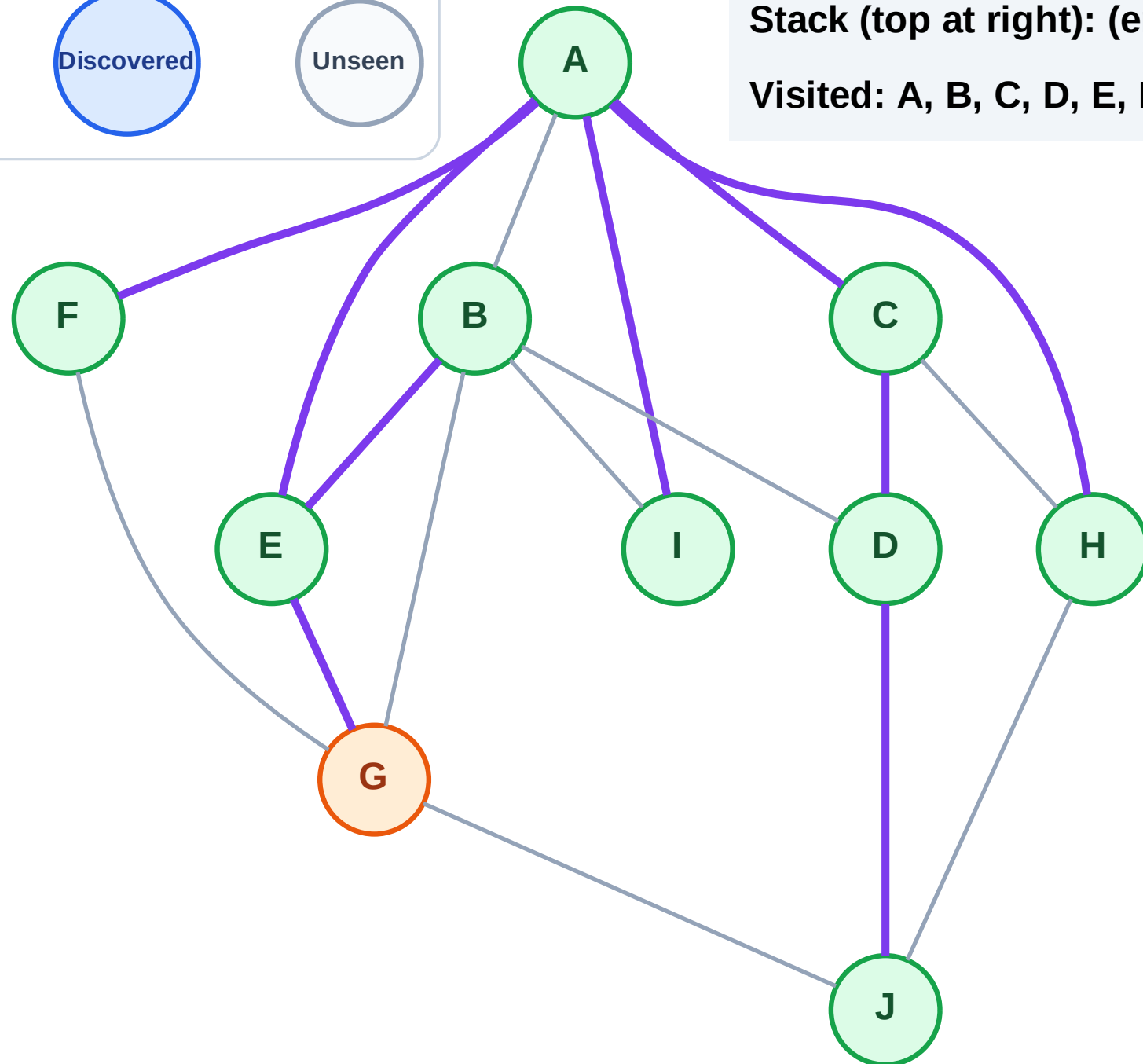
Step 20: Process node G

Legend



Stack (top at right): (empty)

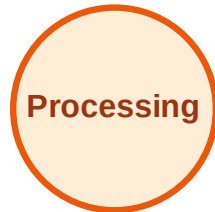
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

